



**Technical Campus**

योग: कर्मसु कौशलम्  
IN PURSUIT OF PERFECTION



**SCHOOL OF ENGINEERING  
AND TECHNOLOGY**

# Process Automation (IOT310)

**By: Dr. Divya Agarwal**

## Course Overview:

- Process and Industrial Automation specialization offers comprehensive knowledge and professional level skills focused on developing and deploying software robots. It starts with the basic concepts of Robotic Process Automation. In the present global scenario, industries are moving towards complete automation. Hence, this course is foundation of engineers who want to specialize in industrial and process automation field.

## Course Objective:

- To understand the Fundamental and basic principles of Robotics Process Automation, Applications in various Industries.
- To install the UiPath Studio and understand difference between different versions
- To use different activities while using excel sheet and automate email related activities.
- To understand benefits of given industrial automation systems, describe their functions and compare characteristics of given automation systems.

## Course Outcomes (CO):

- **CO1** Able to Map and assess some of the business processes that are fit for automation.
- **CO2** Able to connect to UiPath Automation Cloud, download the installer, and set up their own attended automation environment and Build an automation using StudioX.
- **CO3** Able to effectively automate tasks involving use of Excel files, adapt the automation project to dynamic Excel ranges.
- **CO4** Able to apply and map realtime application of industrial automation

- **UNIT I**

❑ **Introduction to process automation:** Definition of automation, Socio economic impacts of automation, Types of Automation, Introduction to Robot Process Automation, How is Automation Driving the Digital Transformation? Role of Automation in Business; Build first automation process with UiPath StudioX

# Syllabus Contents

- **UNIT II**

- **Automation using StudioX:** computational concepts when building an automation project, Identify the tasks that are suitable for automation, breakdown a task and document it in a RobotPath. Introduction to file and folder automation,
- **Introduction to UI Automation:** Introduction to UI automation, recording UI interactions, using the Object Repository, UI automation activities, extracting data from an application, using of different control flow activities and its use

# Syllabus Contents

- **UNIT III**
- **Error Handling:** Use StudioX tools to validate and analyze automation projects for handling and troubleshooting errors, build automations using best practices that increase reusability and readability.
- **Email Automation:** Use actions and resources related to email automation, create automation projects using StudioX email specific activities, create email content using both text and HTML options.
- **Excel Automation:** Automate tasks involving use of Excel files such as usage of cell activities, range activities, pivot activities, chart activities and workbook activities.

# Syllabus Contents



- **UNIT IV**
- **Introduction to Industrial Automation:** Need and benefits of Industrial Automation, Automation Hierarchy, Basic components of Automation systems, description of each component, types of automation systems flexible, fixed and programmable systems
- **The Future Automated Factory:** Future Automated Factory, Human Workers in the Future Automated Factory, The social impact

# Syllabus Contents

- **UNIT II**

- **Automation using StudioX:** computational concepts when building an automation project, Identify the tasks that are suitable for automation, breakdown a task and document it in a RobotPath. Introduction to file and folder automation,
- **Introduction to UI Automation:** Introduction to UI automation, recording UI interactions, using the Object Repository, UI automation activities, extracting data from an application, using of different control flow activities and its use



# ABOUT Ui PATH

- UiPath is an RPA vendor that provides software to help organizations automate their business processes.
- Objective is to automate repetitive and tedious tasks, allowing humans to engage in more creative and inspiring activities.
- With UiPath automation software, one can configure software Robots to mimic human action on the user interface of computer systems.
- Basic components of UiPath RPA are
  - UiPath Studio,
  - UiPath Robot, and
  - UiPath Orchestrator

- UiPath Studio helps users with no coding skills to design Robotic processes in a visual interface.
- It is a flowchart-based modeling tool. Thus, automation is faster and more convenient.
- Multiple people can contribute to the same workflow.
- Presence of a visual signal that points out errors in the model, and a recorder that performs what the user executes, make modeling much easier.

- UiPath Robot runs the processes designed in UiPath Studio.
- It works in both attended (working only on human trigger) and unattended environments (self-trigger and work on their own).
- **Types of Robots:**
  - **Attended**: Operates on the same workstation as a human to help the user accomplish daily tasks. Usually triggered by user events. Cannot start a process from Orchestrator on these type of Robots, and they cannot run under a locked screen.
  - **Unattended**: Can run unattended in virtual environments and can automate any number of processes. This robot is responsible for remote execution, monitoring, scheduling, and providing support for work queues.
  - **Free**: Similar to Unattended Robots, but can be used only for development and testing purposes, not in a production environment.

# UiPath Orchestrator

- UiPath Orchestrator is a web-based platform that runs and manages Robots.
- Capable of deploying multiple Robots, and monitoring and inspecting their activities.
- Orchestrator helps in creating and maintaining the connection between Robots
- Ensures the correct delivery of the packages to Robots
- Helps in managing the queues
- Helps in keeping track of Robot identification
- Stores and indexes the logs to SQL or Elasticsearch
- Behind the scenes, Orchestrator Server uses: IIS Server, SQL Server, Elasticsearch, Kibana

# Computational Concepts in Building Automation Projects with StudioX

- StudioX, a low-code automation platform, leverages several computational concepts to simplify complex automation tasks.
- Doesn't require extensive programming knowledge,
- Understanding following concepts can enhance automation projects:
  1. **Data Structures**
  2. **Conditional Statements**
  3. **Loops**
  4. **Functions**
  5. **Error Handling**
  6. **Data Integration**

## 1. Data Structures:

- **Variables:** StudioX uses variables to store and manipulate data during automation. Different data types include text, numbers, dates, and booleans. Variables hold temporary values used throughout your workflow.
- **Collections:** Arrays and lists hold multiple data items of the same type. They allow you to manage groups of data efficiently within your automation.
- **Data Tables:** StudioX uses data tables to structure and organize large datasets. They provide a spreadsheet-like interface to store and manipulate various data points relevant to your automation.

## 2. Conditional Statements:

- **If/Else Statements:** These control the flow of your automation based on specific conditions. If a condition is true, one set of actions is executed; otherwise, another set follows.
- **Switch Statements:** When dealing with multiple conditions, Switch statements offer a more concise way to choose between different execution paths based on the value of a variable.

## 3. Loops:

- **For Loops:** These repeat a set of actions a specific number of times. They are useful for iterating through data collections or performing repetitive tasks.
- **While Loops:** These continue executing a set of actions as long as a specific condition remains true. They are helpful for tasks requiring repeated checks or actions until a certain criteria is met.



## 4. Functions:

- **Built-in Functions:** StudioX provides a library of pre-built functions for various tasks like data manipulation, string operations, and mathematical calculations. They simplify common tasks and promote code reusability.
- **Custom Functions:** Custom functions can be created to encapsulate complex logic and reuse it across automation projects. This improves modularity and organization.

## 5. Error Handling:

- **Try/Catch Blocks:** These allow to gracefully handle potential errors during automation. "Try" block executes code, and "Catch" block handles any exceptions that occur, preventing automation from crashing.

## 6. Data Integration:

- **API Calls:** StudioX allows to interact with external applications and services through APIs (Application Programming Interfaces). This enables data exchange and automation across different platforms.
- **Database Interactions:** Connect to and manipulate data stored in databases using StudioX. This allows to retrieve, update, and manage data relevant to automation process.

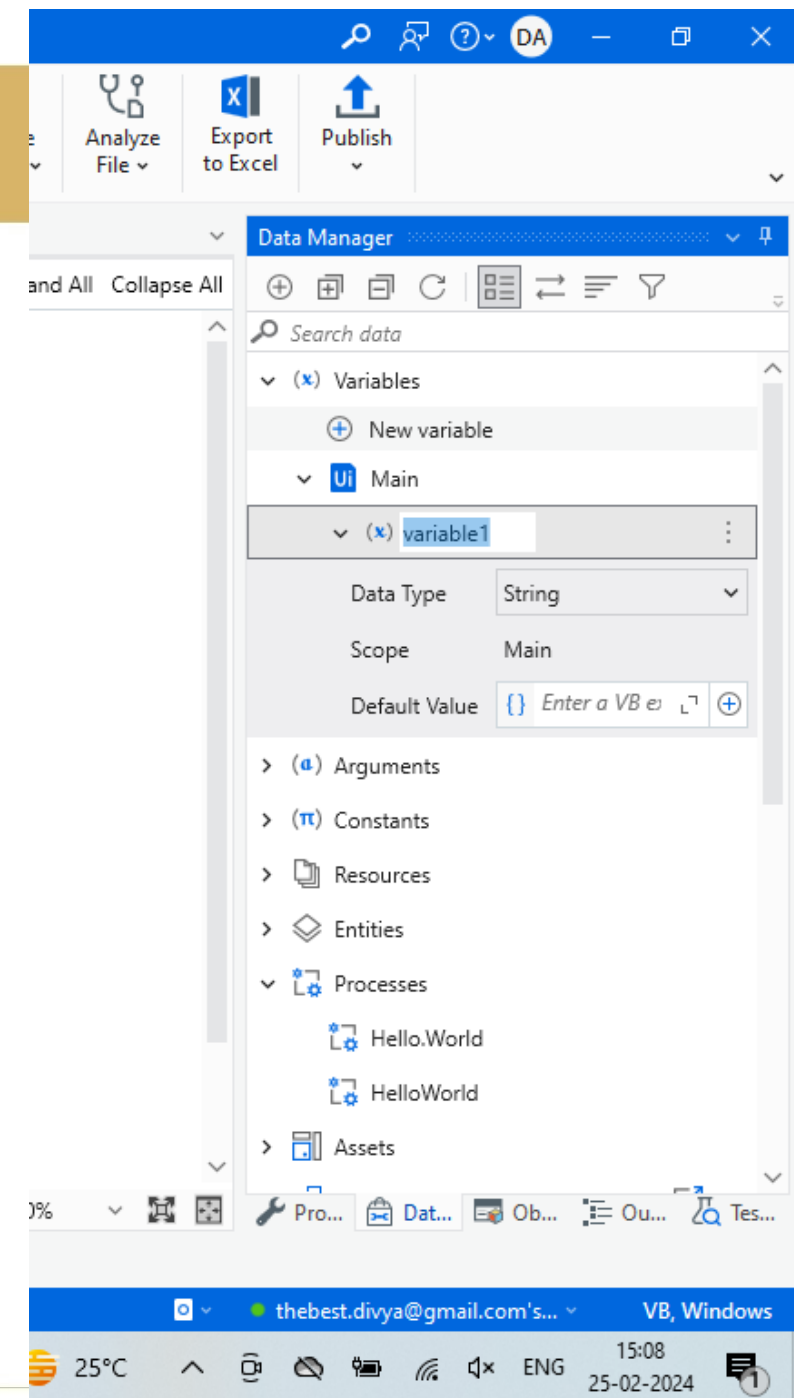
- The data stored within a variable is called a value, and it can be of multiple types.
- Types of variables include:
  - **Boolean** - Has only two possible values: true or false. These variables enables to make decisions, and therefore have a better control over workflow.
  - **Int32** - Used to store numeric information. Some of its uses are to perform equations, comparisons and pass important data.
  - **String** - Stores text information. Used to store any text-based information such as employee names, usernames or any other strings.

- **Object** - Used to store different types of data. Use this type of variable in situations where you need flexibility for the type of data you store in a variable. Data type used in Object variables is compatible with the activities that use the variable.
- **System.Data.DataTable** - Can store big pieces of information, and act as a database or a simple spreadsheet with rows and columns. Can be useful to migrate specific data from a database to another and extract information from a website and store it locally in a spreadsheet, among other uses.
- **Array of [T]** - Enables to store multiple values of the same type.
- **Browse for Types** - Allows to [explore more types](#) that can be relevant to task.

# How to create Variables

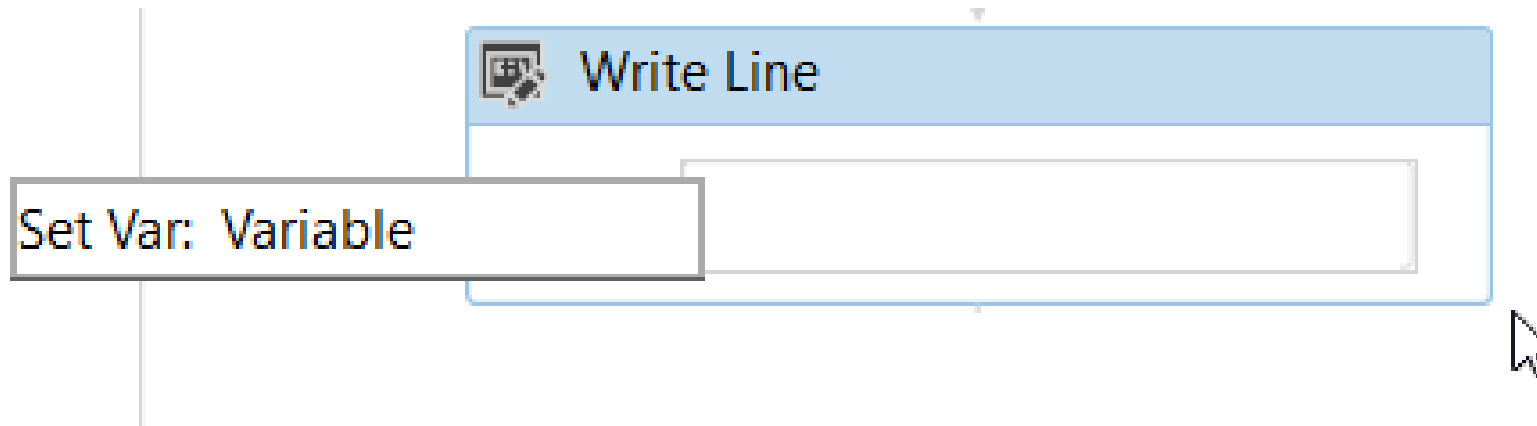
## From the Data Manager

- Open the Data Manager, then select **New > New Variable**.
- Configure the name, data type, scope, and default value for the variable.



# How to create Variables

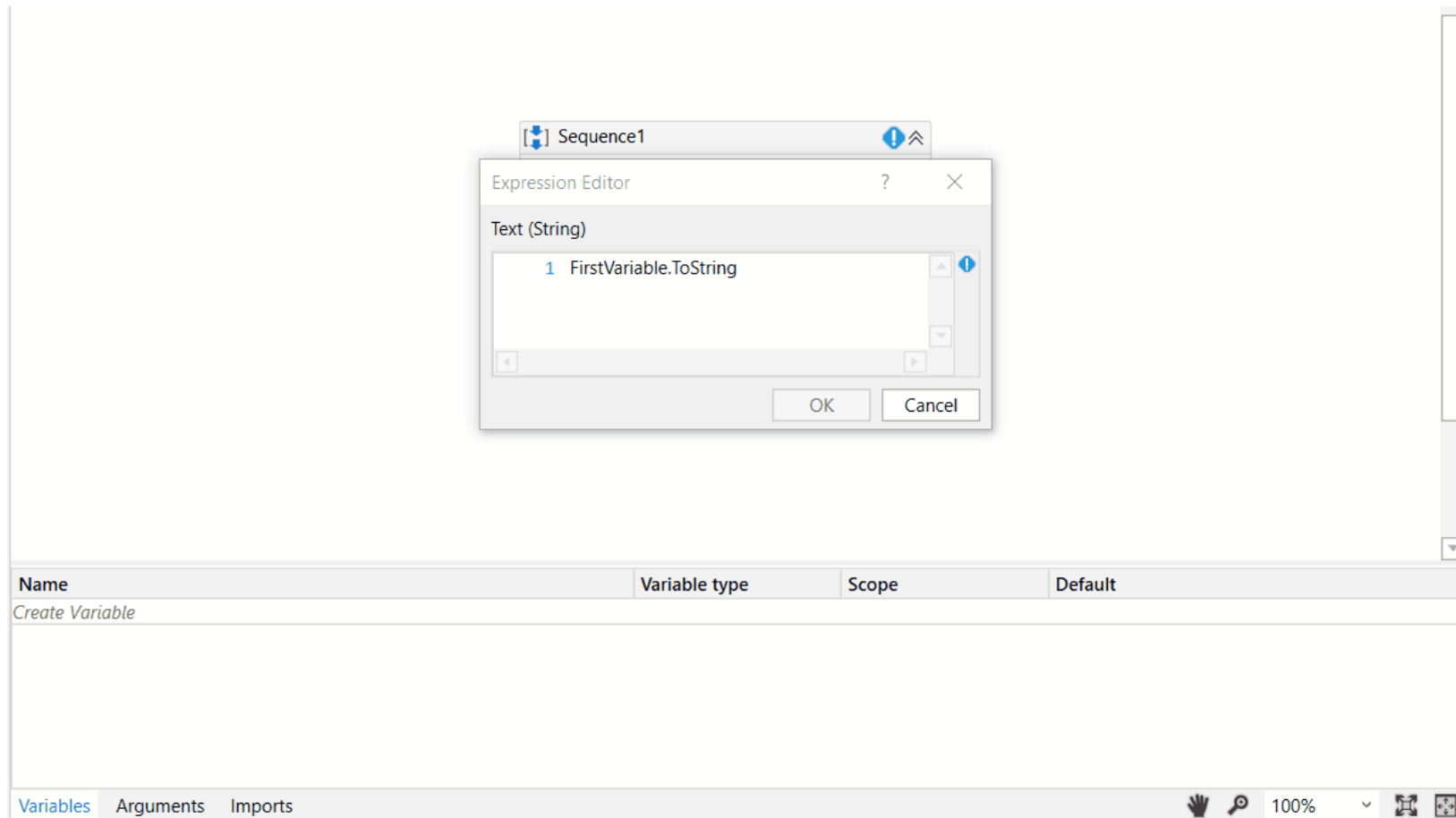
- **From the Body of an Activity**
- From the **Activities** panel, drag an activity to the **Designer** panel. Either right-click a field and select **Create Variable** from the context menu, press Ctrl+K, or select **Create Variable** from the **Plus** menu on the right side of the field. The **Set Var** field is displayed.
- Fill in the name and press Enter. The variable is created and visible in the field. Check its scope and type in the **Data Manager**.



# From Expressions

Variables can be created from expressions directly in an activity input field or the **Expression Editor**:

1. Select a part of the expression and press Ctrl+K. The **Set Var** field is displayed.
2. Fill in the name and press Enter. The variable is created. Check its scope and type in the **Data Manager**.



- Activities are the basic building blocks of an automation project, each representing one or more manual tasks that StudioX can automate.
- A sequence of related, interconnected activities makes up an automation project.
- When considering automating a repetitive process, first break down that process into each of the individual manual steps, and then identify the activities in StudioX that automate each of those steps.
- For example, let's say that you want to automate the following workflow:
  - Open a spreadsheet.
  - Open a business application
  - Copy the data from a cell in the spreadsheet.
  - Paste the copied data into a field in a business application.
  - Click a button in the application to submit the data.

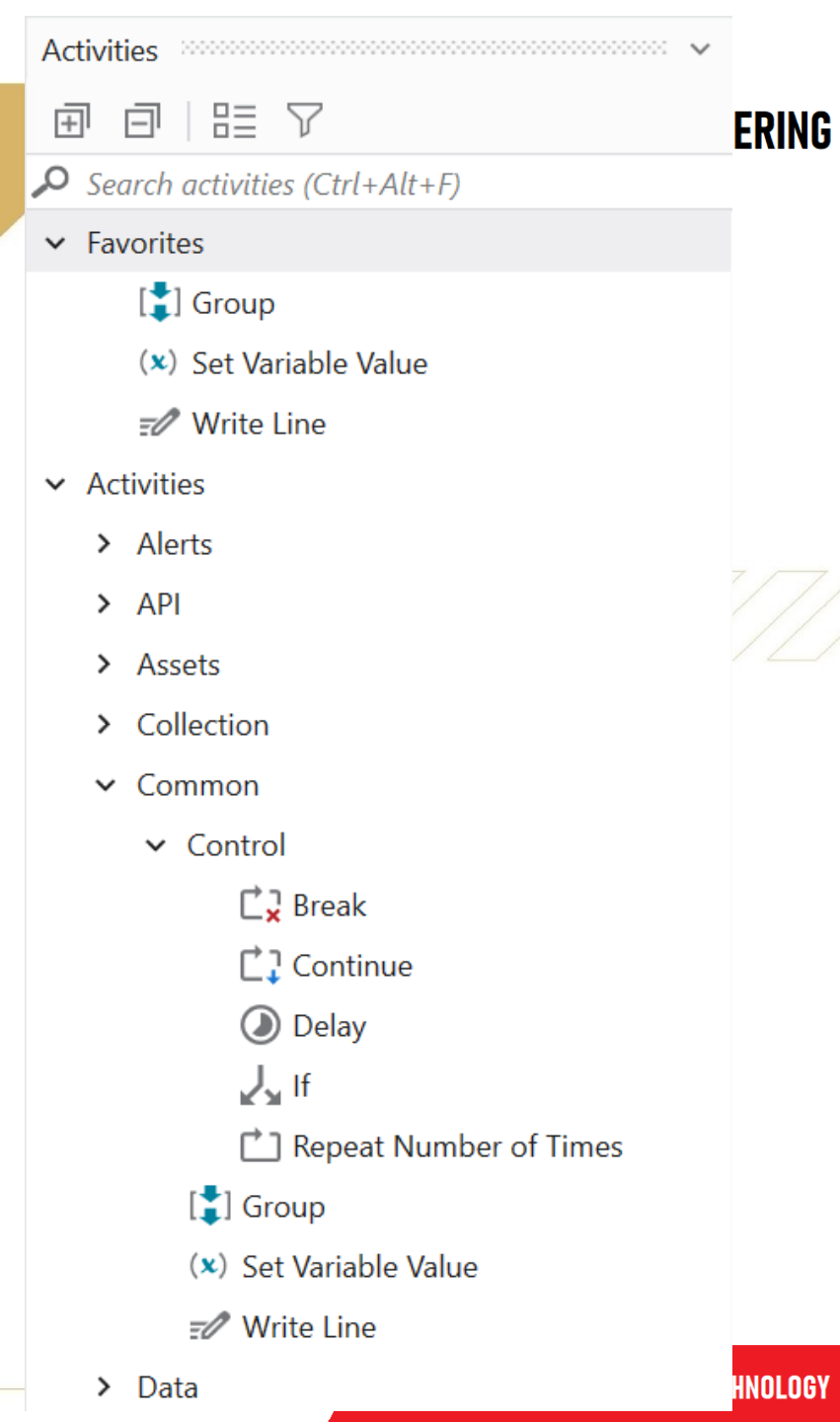


- Activities for automating a process while creating a project are:
- **Use Excel File** to indicate which spreadsheet you want to use.
- **Use Application/Browser** to indicate which application you want to use.
- **Type Into** to select the input field in the business application where to copy the data as well as the cell from which you want to copy the data.
- **Click** to indicate which button to click to submit the data in the business application.

# Activities Panel

- Activities panel shows available activities that can be added to the current project.
- In Windows and cross-platform projects, you can see both installed and installable activities, while in Windows - Legacy projects only installed activities are visible.
- Installed refers to activities packages that are already part of a project, while installable refers to packages that are available to install from the **Official** and **Orchestrator Tenant** feeds.
- To add an activity to your automation, double-click it or drag and drop it to the desired place in the Designer panel.
- Adding an installable activity installs its corresponding activities package in the project.
- Hovering over an activity in the panel displays the package it belongs to and its description.
- To add an activity to the list of favorites, right-click it and select **Add to favorites**.

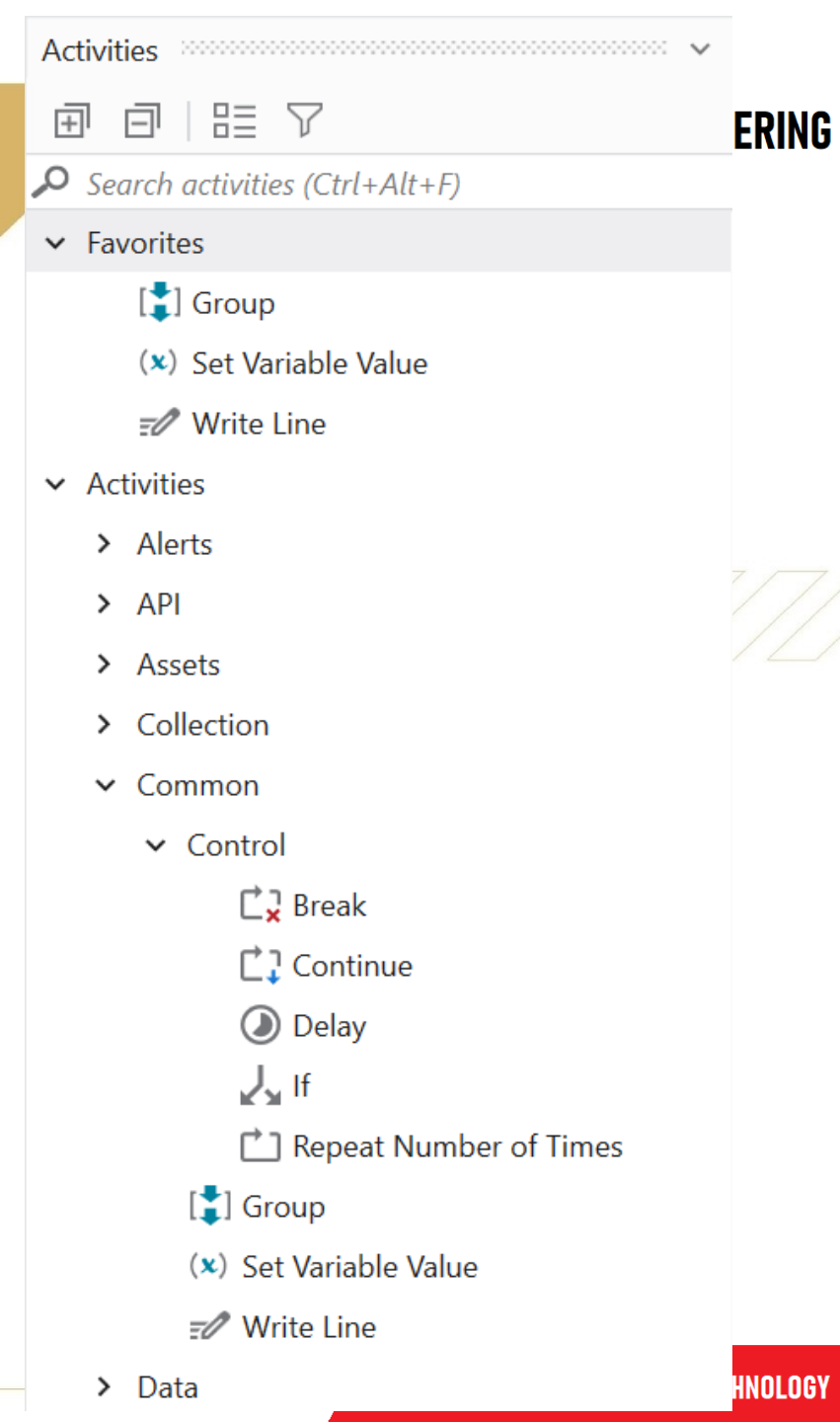
Dr. Divya Agarwal



# How to customize Activities Panel

- Click the **Group by** button at the top of the panel to group available activities by the category they belong to or by the package they belong to.
- Click the **Filter by** button to filter activities based on the options below.
- **For Windows and cross-platform projects**
- **Filter by availability:**
  - **Installed** - Show activities that are part of packages installed in your project (this filter cannot be deactivated)
  - **Available** - Show discovered activities that you can add to your project
  - **Windows** - Show activities that are only available in Windows projects (this filter is only available in cross-platform projects)
- **Filter by type:**
  - **Activities** - Show activities
  - **Triggers** - Show triggers
- **Filter by category:**
  - **Developer** - Show activities designed for the Studio profile.
- For example, if you are trying to view all Excel activities, you can select **Group by Packages** and deselect **Group by Category**.

Dr. Divya Agarwal



ERING

NOLOGY

# Container Activities

- Define a scope or conditions to be met for execution of activities added inside them.
- Can be described as a parent activity that contains one or more child activities.
- Examples of container activities include:
  - [Resource activities](#) - Add an application resource to the automation and make that resource available to the activities added inside them.
  - [For Each activities](#) - Define a collection of items and repeat the activities added inside them once for each of those items.
  - [Repeat number of times](#) - Similar to a For Each, but it repeats the activities added inside it for a specified number of times.
  - [If](#), [Switch](#), and [Check App State](#) - Evaluate a condition and determine the flow of the automation by executing specific activities added inside them based on the result of the evaluation.
  - [Wait for Download](#) - Detects a file download initiated by the activities added inside it and makes the file available in the automation.

# Resource Activities

- [Use Excel File](#) - Use a specified Excel file. Data in Excel file is available to all activities added inside this activity. When you configure child activities, you can select individual cells, ranges, or sheets in file directly from StudioX.
- [Use Desktop Outlook App](#), [Use Outlook 365](#), and [Use Gmail](#) add a desktop Outlook, Outlook 365, and Gmail account to automation. Emails in account are available to all activities added inside this activity. When you configure child activities, you can select individual folders, messages, or message fields directly from StudioX.
- [Use Word File](#) - Use a specified Word document. Add activities to perform on file inside this activity.
- [Use Application/Browser](#) - Use a specified desktop application or web browser page. Add activities to perform in application or web browser inside this activity.
- [Use PowerPoint Presentation](#) - Use a specified PowerPoint file. Add the activities to perform on the file inside this activity.
- [Use OneDrive & SharePoint](#) - Select an account to use for OneDrive and SharePoint automation. Add the related activities inside this activity.

# Resource Activities

- You can add as many resources as your automation requires.
- For activities that use more than one application, add the resources one inside the other.
- For example:
  - To add activities that use both data from an Excel file and emails from an Outlook account, add a Use Excel File activity inside a Use Desktop Outlook App activity, and then add the common activities inside Use Excel File.
  - To add activities that use both data from a web page and a Word document, add a Use Application/Browser activity inside a Use Word File activity, and then add the common activities inside Use Application/Browser.

# Repeating one or more activities in automation project

- To iterate through items, add one of the available **For Each** activities in which you define the collection, and then add the activities to repeat inside the For Each.
- StudioX comes with the following **For Each** activities:
  - [For Each Excel Row](#) - Repeat one or more activities for each row in an Excel table, range, or sheet.
    - The current item option is CurrentRow. For some activities, you must also select which column to use from the current row.
  - [For Each Excel Sheet](#) - Repeat one or more activities for each sheet in an Excel file.
    - The current item option is CurrentSheet.
  - [For Each Email](#) - Repeat one or more activities for each message in an Outlook folder, Gmail label, or in a selection of messages.
    - The current item option is CurrentMail. For some activities, you must also select which field to use from the current mail: Subject, Body, Body as HTML, Bcc, Cc, To, From, or Priority.

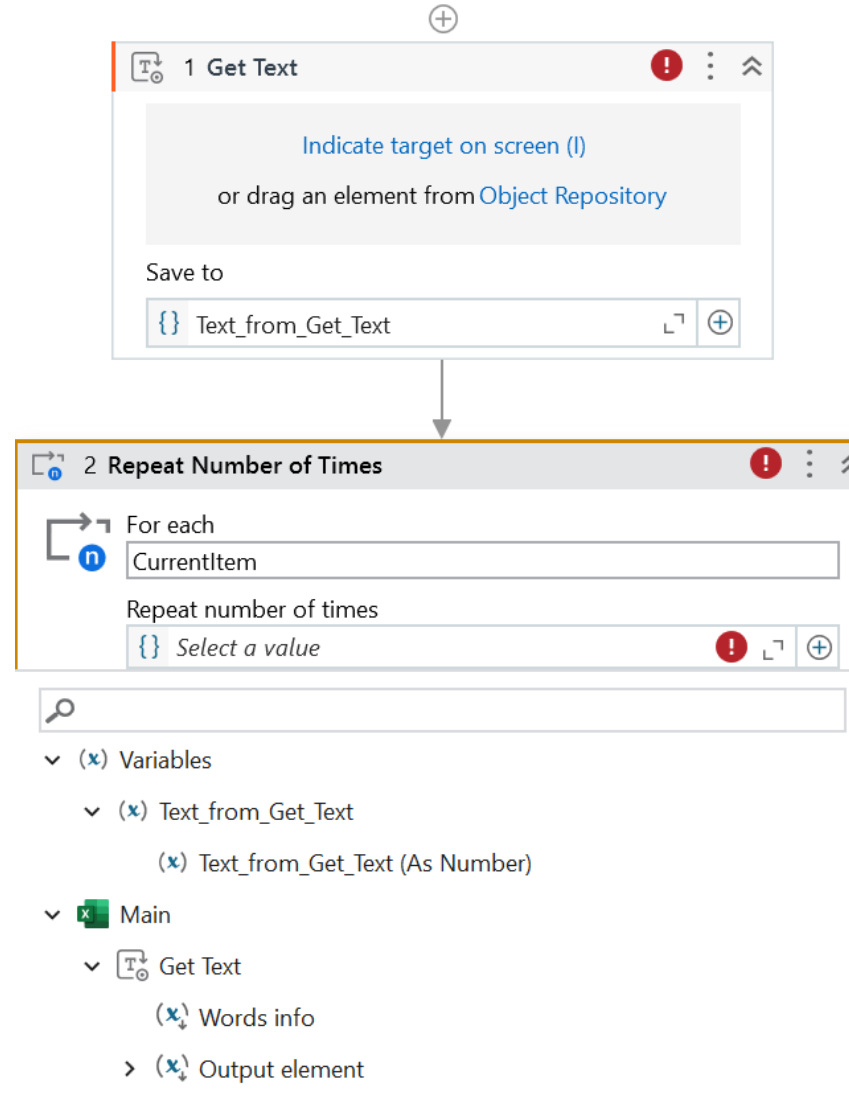


# Repeating one or more activities in automation project

- [For Each File in Folder](#) - Repeat one or more activities for each file in a folder on your computer.
  - The current item option is **CurrentFile**. You must also select which file property to use from the current file: Size, Name, Full Name (includes full path), Folder, Last modified date, Last accessed date, Created date, IsReadOnly, or Size in KB.
- [For Each Folder in Folder](#) - Repeat one or more activities for each folder inside a specified folder on your computer.
  - The current item option is **CurrentFolder**. For some activities, you must also select which folder property to use from the current folder: Name, Full Path, or Last modified date.
  - To make it easier to identify when configuring activities added inside a For Each, you can rename the current item option using a name that describes the items you are iterating through.



# Repeating one or more activities in automation project



# Task Identification:

- Begin by identifying tasks that are repetitive, rule-based, time-consuming, and prone to human error.
- These are prime candidates for automation.
- Look for tasks that involve data entry, data manipulation, file processing, report generation, or interactions with software applications.

# Task Identification:

- **Repetitive and Rule-Based:** Tasks with clear, defined steps and predictable outcomes are ideal for automation. RPA excels at following rules and executing actions consistently without human intervention.
- **Time-Consuming and Manual:** Tasks that consume significant human effort and time, especially for large volumes of data, are prime candidates for automation. RPA can free up human resources for more valuable tasks.
- **Error-Prone:** Tasks susceptible to human error due to fatigue, distraction, or data entry mistakes can benefit from automation. RPA ensures accuracy and consistency in repetitive tasks.
- **Data-Driven:** Tasks involving data manipulation, extraction, or migration are well-suited for automation. RPA can handle large datasets efficiently and accurately.
- **Stable and Standardized:** Processes with minimal variation or change are easier to automate. Frequent changes may require frequent updates to the automation script.

# Task Analysis:

- Once you've identified a task for automation, analyze it thoroughly to understand its scope, inputs, outputs, dependencies, and exceptions.
- Break down the task into smaller, more manageable steps or subtasks.
- This decomposition helps in understanding the intricacies of the process and identifying automation opportunities at a granular level.

# Breaking Down a Task into a RobotPath

- **Define the Task Scope:** Clearly define the beginning and end points of the task you want to automate. What triggers the task, and what are the desired outcomes?
- **Identify Subtasks:** Break down the main task into smaller, manageable subtasks. List each step in the process chronologically.
- **Determine Data Inputs and Outputs:** Identify the data required for each subtask and what data is produced as an output.
- **Choose Automation Activities:** For each subtask, select the appropriate RPA activities from your chosen platform (e.g., StudioX) that correspond to the required actions.

# Breaking Down a Task into a RobotPath

- **Design the Flowchart:** Use a flowcharting tool or a simple diagram to visually represent the sequence of subtasks and activities. Connect them with arrows to show the flow of data and execution.
- **Document Each Activity:** Describe each RPA activity in detail, including its parameters, expected input, and output. Add screenshots or detailed instructions if necessary.
- **Define Exceptions and Error Handling:** Outline potential exceptions or errors that might occur during automation and specify how your RobotPath will handle them.

# Documentation in RobotPath:

- RobotPath is a visual representation or documentation of the automation process, often created using flowcharts or diagrams. It outlines the sequence of actions to be performed by the automation robot.
- Start by creating a flowchart or diagram representing the high-level steps of the task. Use symbols to represent actions, decisions, loops, and connections between steps.
- Each step in the RobotPath should correspond to a specific action or decision point in the automation process. Clearly label each step to indicate the action to be performed.
- Include annotations or comments to provide additional context or details about each step. This helps in clarifying the purpose or rationale behind certain actions.
- Use branching and looping constructs to represent conditional logic or repetitive actions within the automation process.
- Consider including error-handling pathways to address potential exceptions or failure scenarios encountered during automation execution.

- **Task:** Downloading daily sales reports from a website and uploading them to a CRM system



**Task:** Downloading daily sales reports from a website and uploading them to a CRM system

- **Open Web Browser:** Launch the website with the sales reports.
- **Login:** Enter login credentials using predefined variables.
- **Navigate to Reports Section:** Click specific buttons or links to reach the desired report page.
- **Download Report:** Select the appropriate report and trigger the download action.
- **Move Downloaded File:** Move the downloaded file to a designated folder on the local machine.
- **Open CRM System:** Launch the CRM software using specific commands.
- **Navigate to Upload Section:** Locate the specific section for uploading sales data.
- **Upload Downloaded File:** Select the downloaded file from the local folder and upload it to the CRM system.
- **Log Out and Close Applications:** Close the web browser and CRM system after successful upload.

# Validation and Review:

- Once the RobotPath is documented, validate it against the original task and ensure that it accurately captures all the necessary steps and logic required for automation.
- Review the RobotPath with relevant stakeholders, including process owners, subject matter experts, and automation developers. Gather feedback and incorporate any necessary revisions or adjustments based on their input.

# Implementation in Automation Tool:

- After finalizing the RobotPath, translate it into executable automation logic using an automation tool such as UiPath StudioX.
- Map each step in the RobotPath to corresponding actions or activities in the automation tool's workflow designer.
- Configure input parameters, variables, conditions, and error-handling mechanisms as per the documented logic in the RobotPath.
- Test the automation workflow to ensure that it accurately replicates the steps outlined in the RobotPath and achieves the desired outcomes.

# Benefits of Using RobotPath:

- **Clear and Concise Documentation:** Makes the automation process transparent and understandable for both developers and users.
- **Efficient Development:** Provides a structured approach for building and testing RPA solutions.
- **Repeatability and Scalability:** Ensures consistency in automation across different environments and allows for easier scaling to handle increased workload.

# What's File and Folder Automation?

- File and Folder Automation involves automating tasks you do with your files and folders on your computer, like creating, copying, or renaming them.
- StudioX offers a number of activities that help users automate tasks having to do with files and folders.

# What's File and Folder Automation?

Copy

Create

Delete

Move

Rename



Browse for...

Check if an  
item exists

Iterate  
through...

Get info

Work with  
archives

# What's File and Folder Automation?



**Copy**

## Copy



Copying a file or a folder corresponds to the Copy File and Copy Folder activities.



**Create**

## Create



We can create a new file or folder with the Create File and Create Folder activities.



## Delete

Likewise, we can delete a file or folder with the Delete File and Delete Folder activities.



**Delete**



## Move

The Move File and Move Folder activities allow us to move a file or a folder, giving us the option to overwrite duplicates.



**Move**

# What's File and Folder Automation?

## Rename

With the Rename File and Rename Folder actions, we can rename a file or a folder, being able to choose between keeping the extension or not, in the case of a file.



### Rename



### Browse for...



## Browse for...



The Select File action opens a dialogue box during run time which allows us to choose the file we want to work with. Browse for Folder works similarly, but for folders.

## Check if an item exists



File Exists and Folder Exists are two activities that check whether a file or folder exists. We provide the file or folder path and the answer to this question is stored in a variable that we define within the activity. Later on, this variable can be used in other activities: e.g. as a condition in an If activity - if the file exists, then skip the current file.

### Check if a item exist





# What's File and Folder Automation?

## Iterate through...



For Each File in Folder iterates through all the files in a folder, while For Each Folder in Folder does that for all the folders in a folder.

## Get Info



With the Get File Info and Get Folder Info actions, we can obtain the information about a file or a folder and store it in a variable, similar to the File Exists and Folder Exists activities.

## Get info

## Work with archives

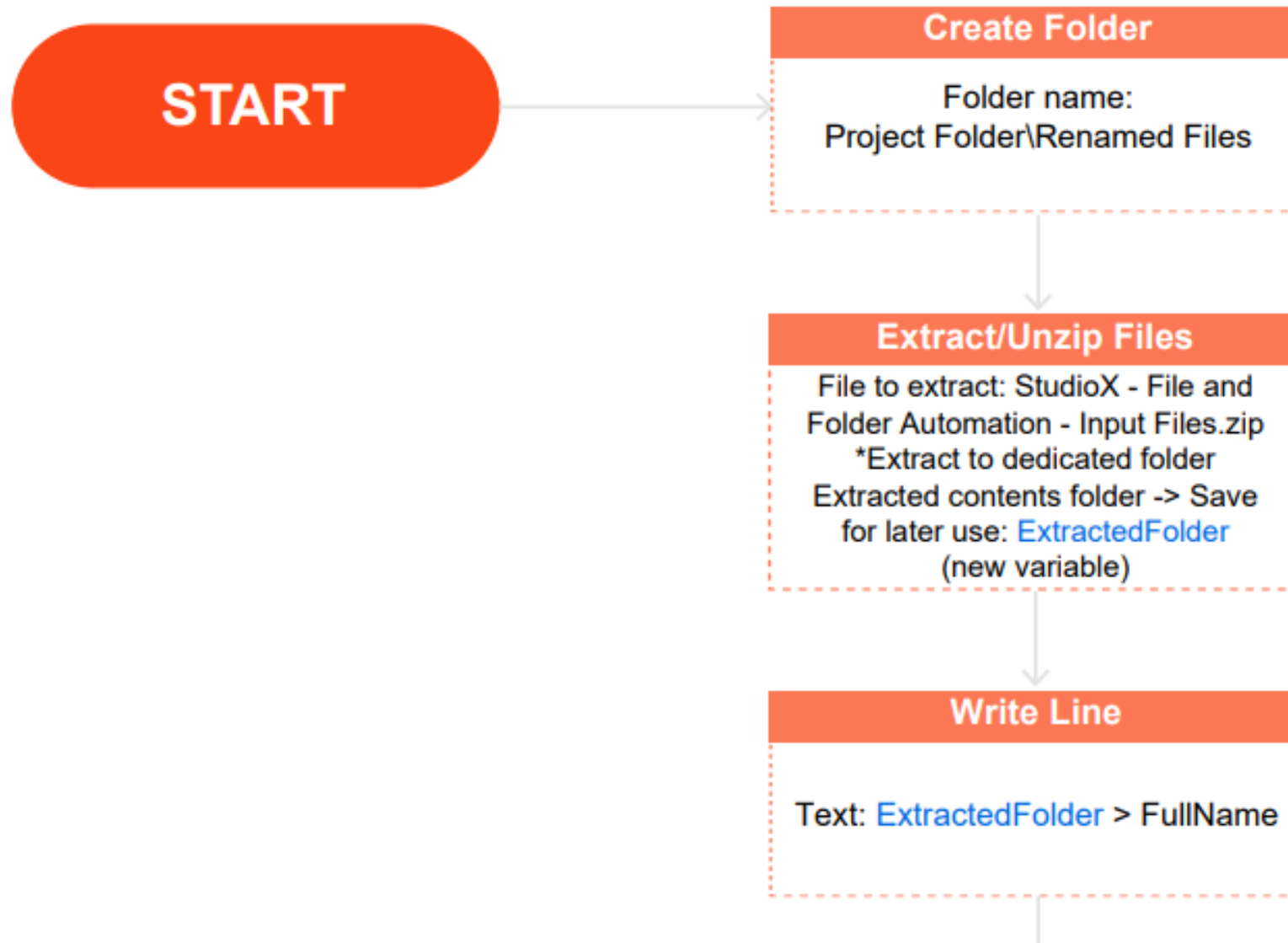


The **Compress/Zip Files** action enables us to add files or folders to a compressed file.

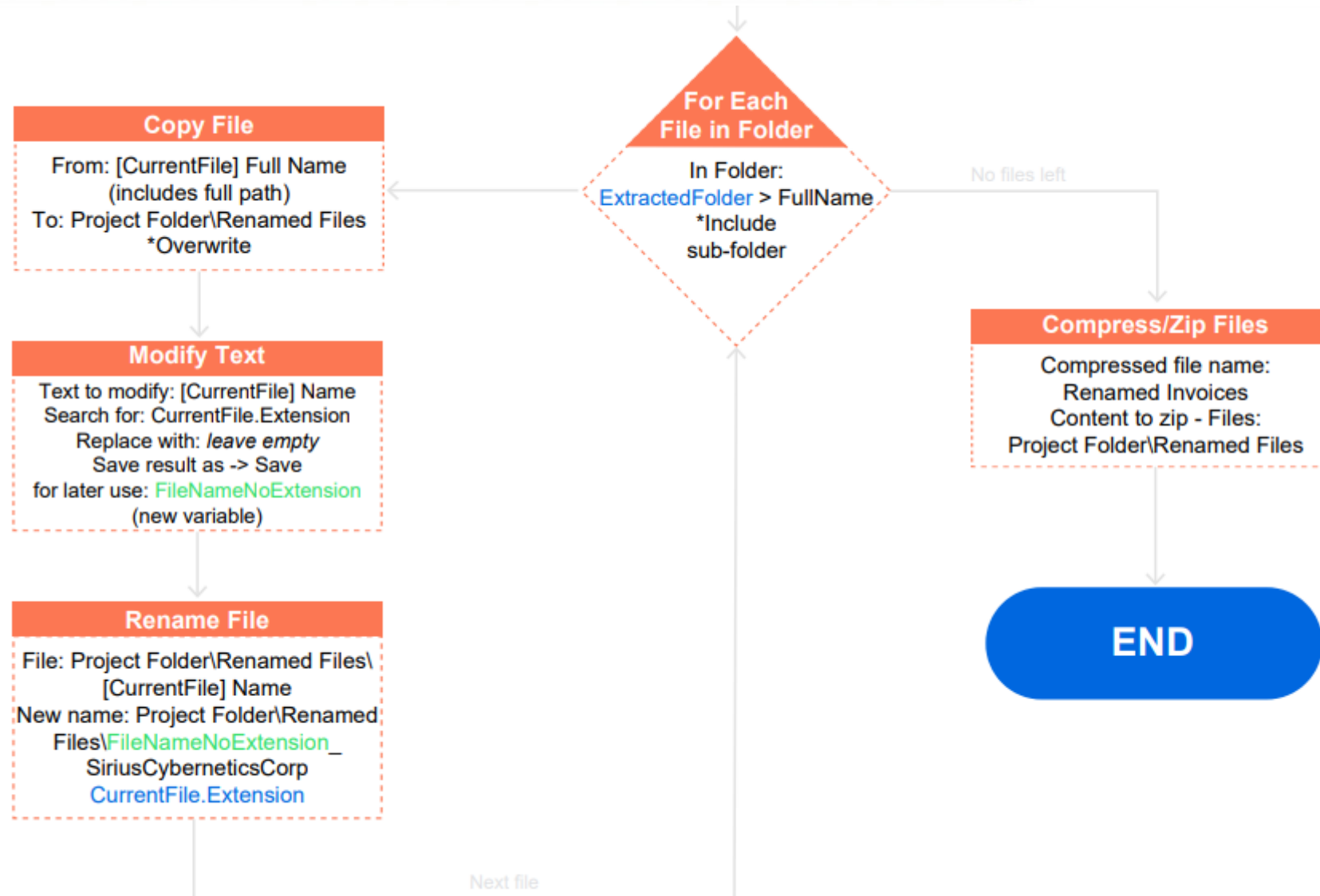
The **Extract/Unzip Files** action helps extract all contents of a compressed file to a specified folder.

## Work with archives

# What's File and Folder Automation?



# What's File and Folder Automation?



# Steps for automating file and folder operations in UiPath StudioX

1. **Open UiPath StudioX:** Launch UiPath StudioX on your computer.
2. **Create a New Project:** Click on "New Project" to create a new automation project.
3. **Select Project Type:** Choose the "Process" project type.
4. **Name the Project:** Give your project a meaningful name related to the file and folder automation task you want to perform.
5. **Start with a Blank Project:** Select the "Blank" template to start with an empty project.
6. **Add a Sequence:** In the Activities Panel on the left, drag and drop a "Sequence" activity onto the design canvas.

# Steps for automating file and folder operations in UiPath StudioX

## 7. Add Activities: Some common activities are:

- **Get Files and Folders:** Drag the "Get Files and Folders" activity to retrieve a list of files and folders from a specified directory.
- **Create Folder:** Use the "Create Folder" activity to create a new folder in a specified directory.
- **Copy File:** Drag the "Copy File" activity to copy a file from one location to another.
- **Move File:** Use the "Move File" activity to move a file from one location to another.
- **Delete File:** Drag the "Delete File" activity to delete a file from a specified location.
- **Rename File:** Use the "Rename File" activity to rename a file.
- **Write Text File:** Drag the "Write Text File" activity to create or overwrite a text file with specified content.
- **Append Line:** Use the "Append Line" activity to add a new line of text to an existing text file.

# Steps for automating file and folder operations in UiPath StudioX

## 8. Configure Activity Properties:

- For each activity added to the sequence, configure its properties by clicking on the activity and setting the required parameters in the Properties Panel on the right.
- Specify input parameters such as file paths, folder paths, file names, etc., based on your automation requirements.

## 9. Add Conditions and Loops (Optional):

- If your automation workflow requires conditional logic or repetitive actions, you can add conditions and loops using activities like "If" and "For Each".
- Drag these activities onto the design canvas and configure their properties to define the conditions or iterations.

# Steps for automating file and folder operations in UiPath StudioX

## 10. Test Your Automation:

- Before running your automation, it's essential to test it thoroughly to ensure that it behaves as expected.
- Click on the "Run" button in the toolbar to execute your automation sequence.
- Monitor the execution in the Output Panel at the bottom of the screen to check for any errors or unexpected behavior.

## 11. Debug and Troubleshoot (If Needed):

- If your automation encounters errors or unexpected behavior, use debugging tools such as breakpoints and step-by-step execution to identify and troubleshoot issues.
- Make necessary adjustments to your automation sequence based on debugging findings.

## 12. Save and Publish:

- Once you're satisfied with your automation, save your project by clicking on the "Save" button in the toolbar.
- You can also publish your automation project to UiPath Assistant or Orchestrator for centralized management and execution.

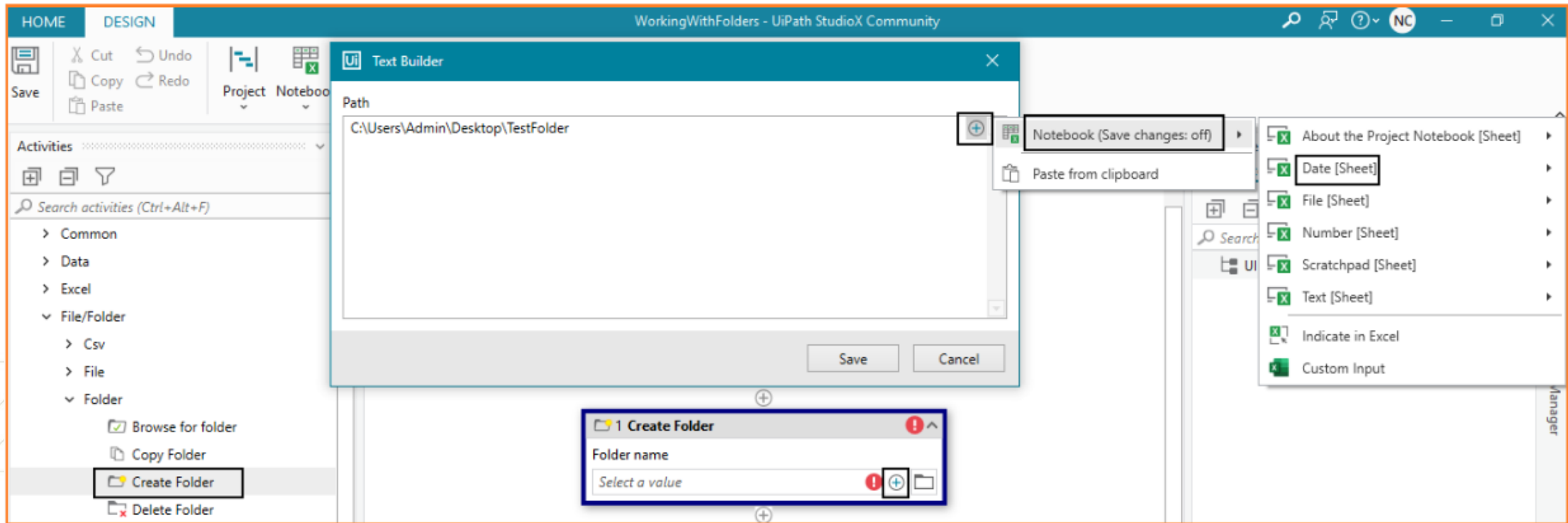
# File and Folder Automation

**Question 1: Create dynamic folder with today's date**



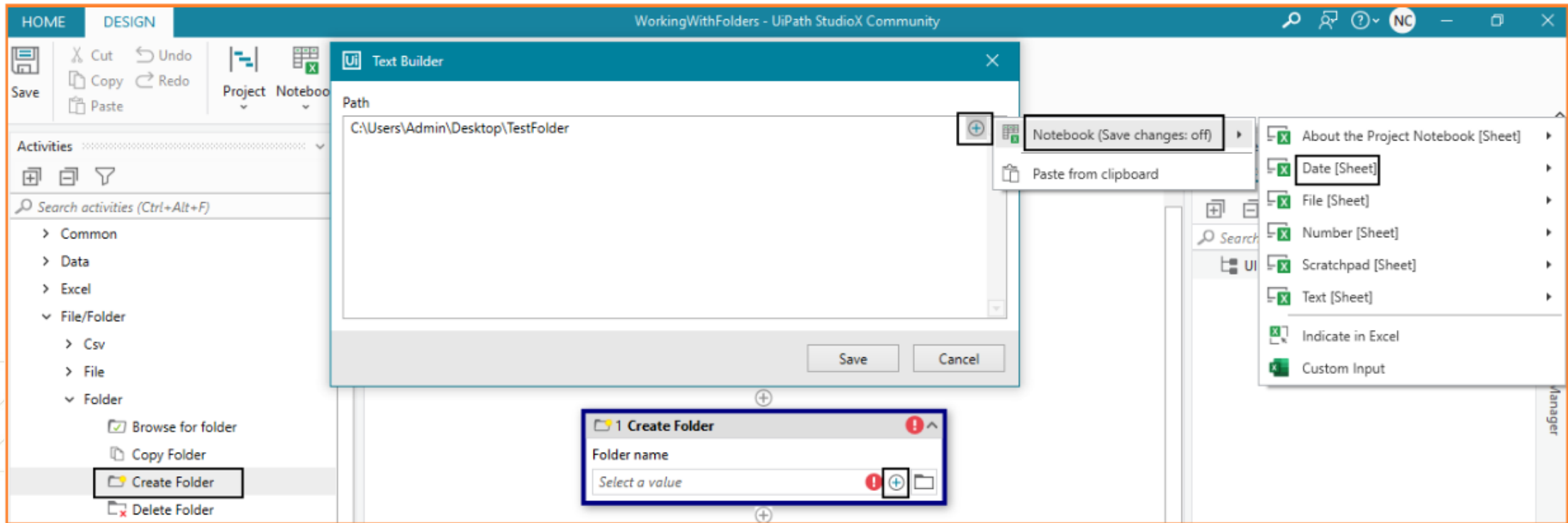
# Create dynamic folder with today's date

- Use 'Create Folder' activity to create the folder. Mention folder path and name in 'Folder name' field.



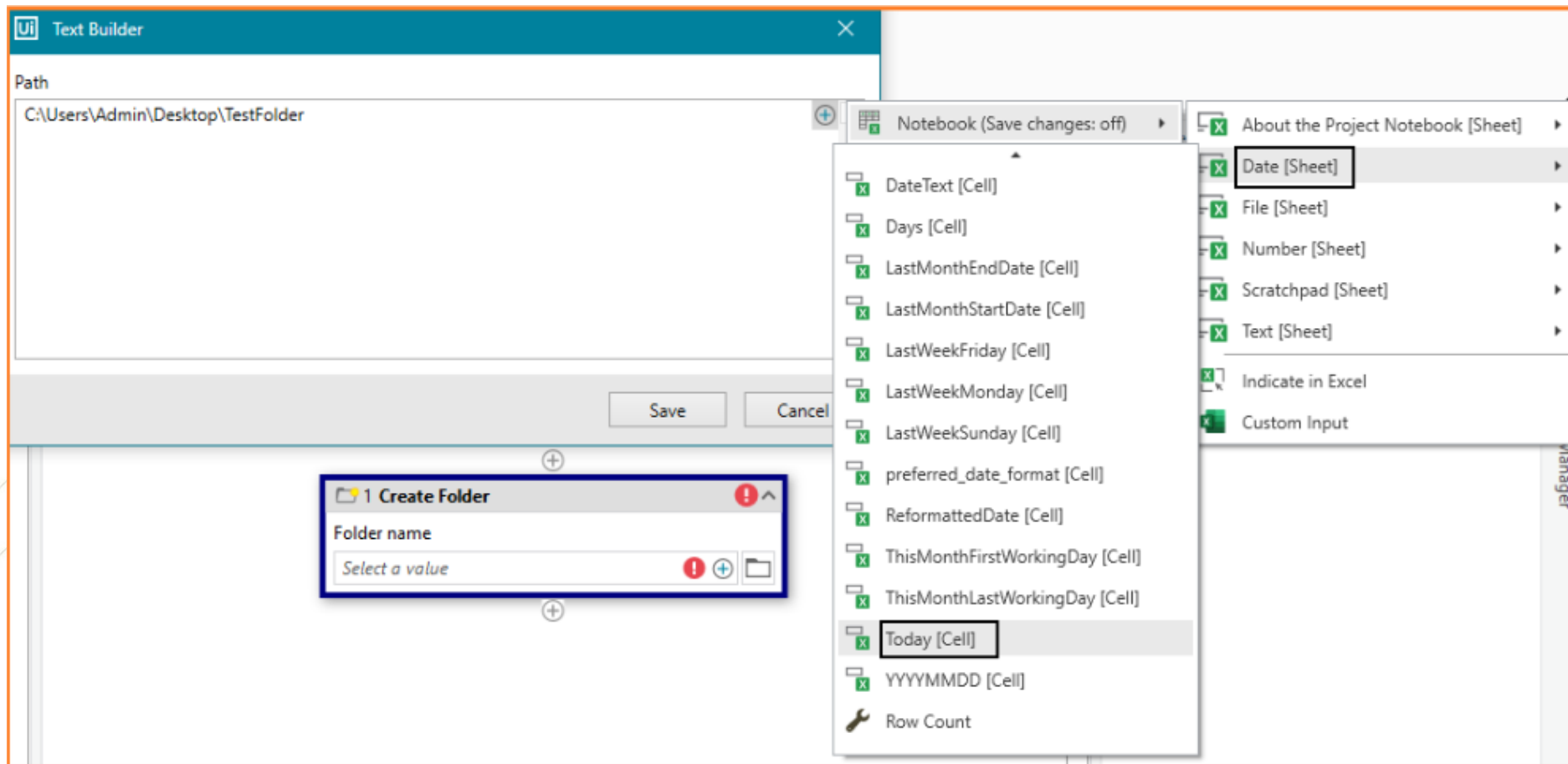
# Create dynamic folder with today's date

- Next, Click (+) in Text builder and choose Notebook.



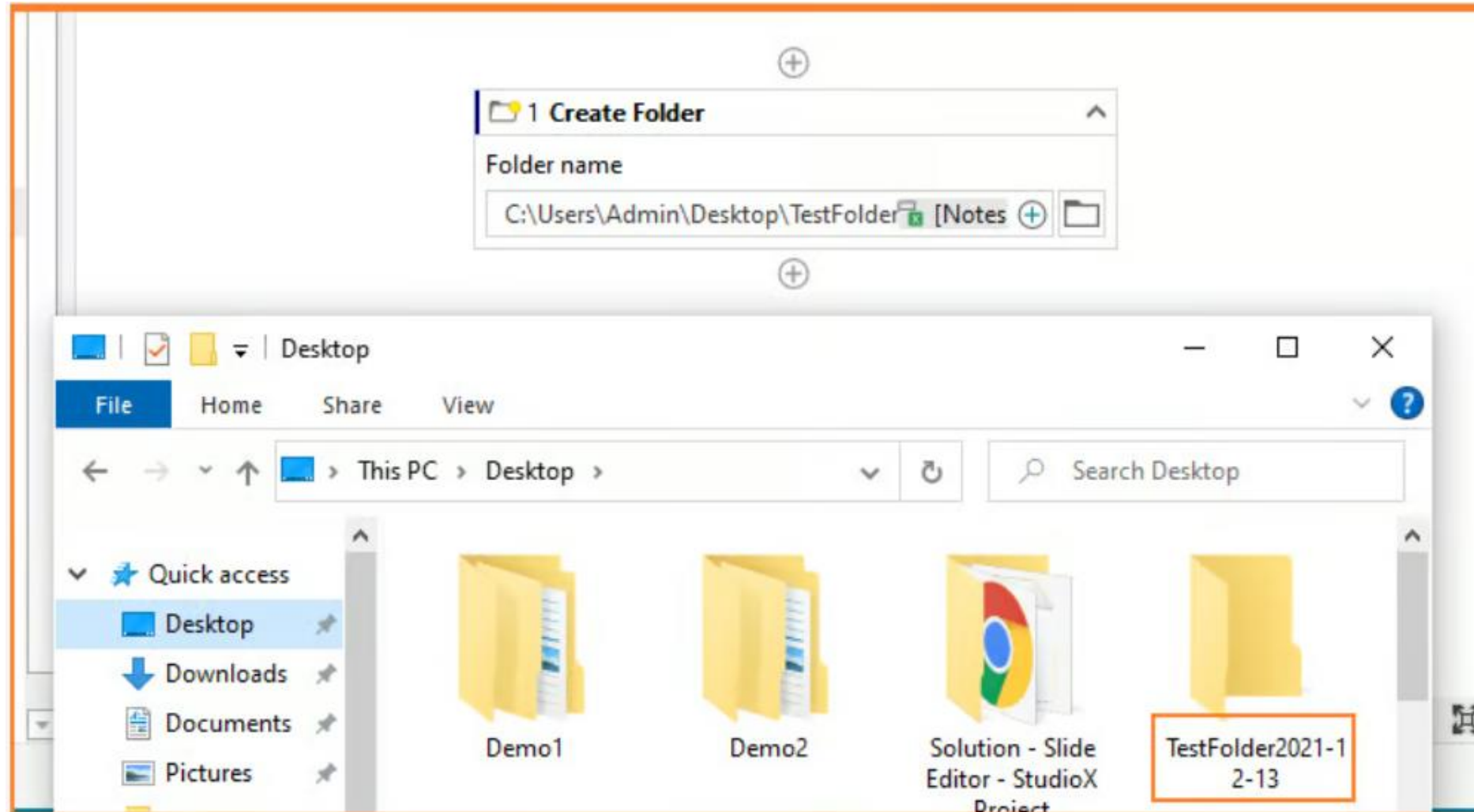
# Create dynamic folder with today's date

- To get today's date, use Project Notebook.  
Read the particular cell of Notebook and add it to the folder path.



# Create dynamic folder with today's date

- Finally, we have a complete path that includes destination path, folder name, and current date. Save and run the automation.

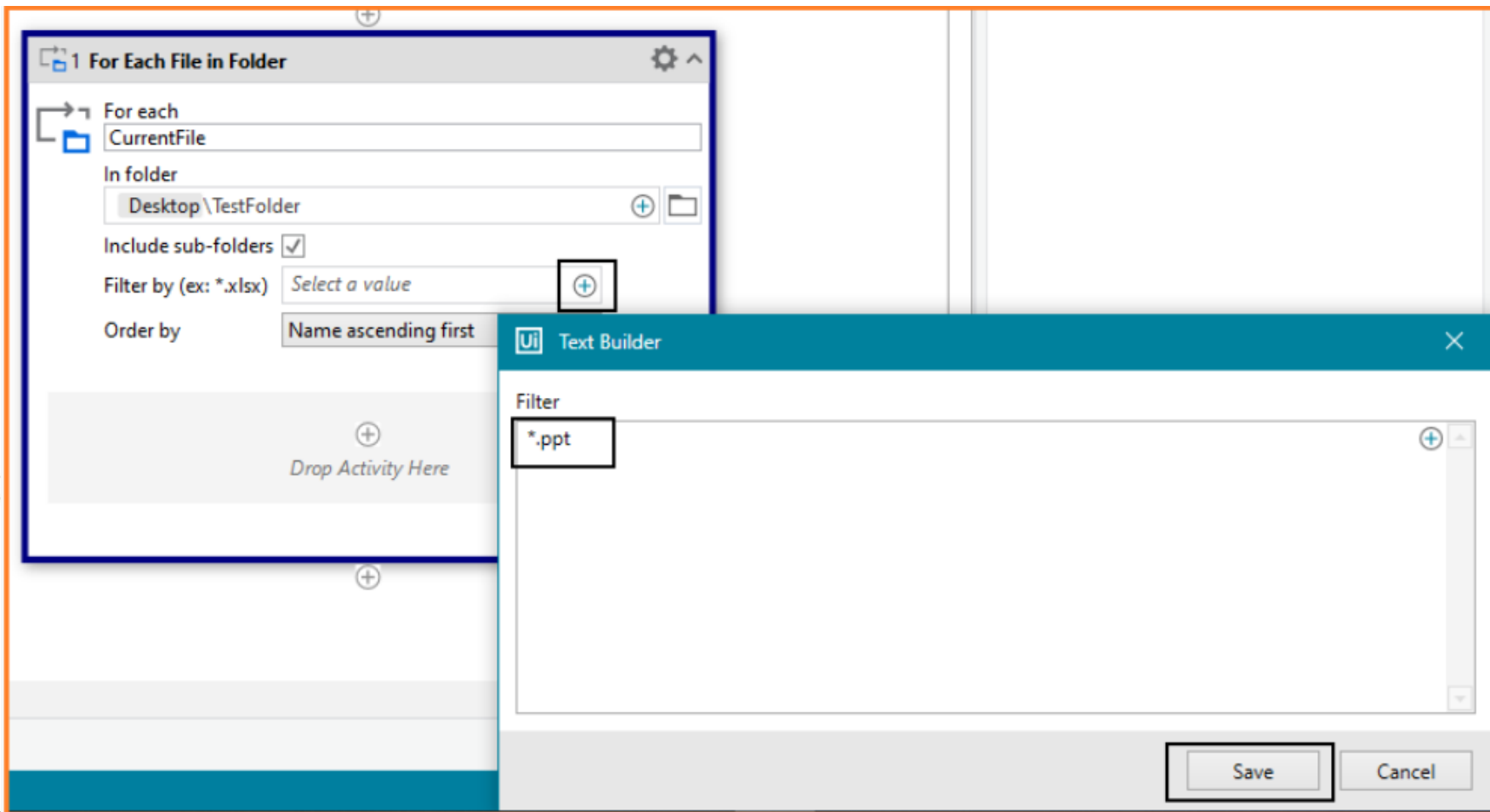


# File and Folder Automation

**Question 2: Filter files with specific file extension  
from a folder**

# Filter files with specific file extension from a folder

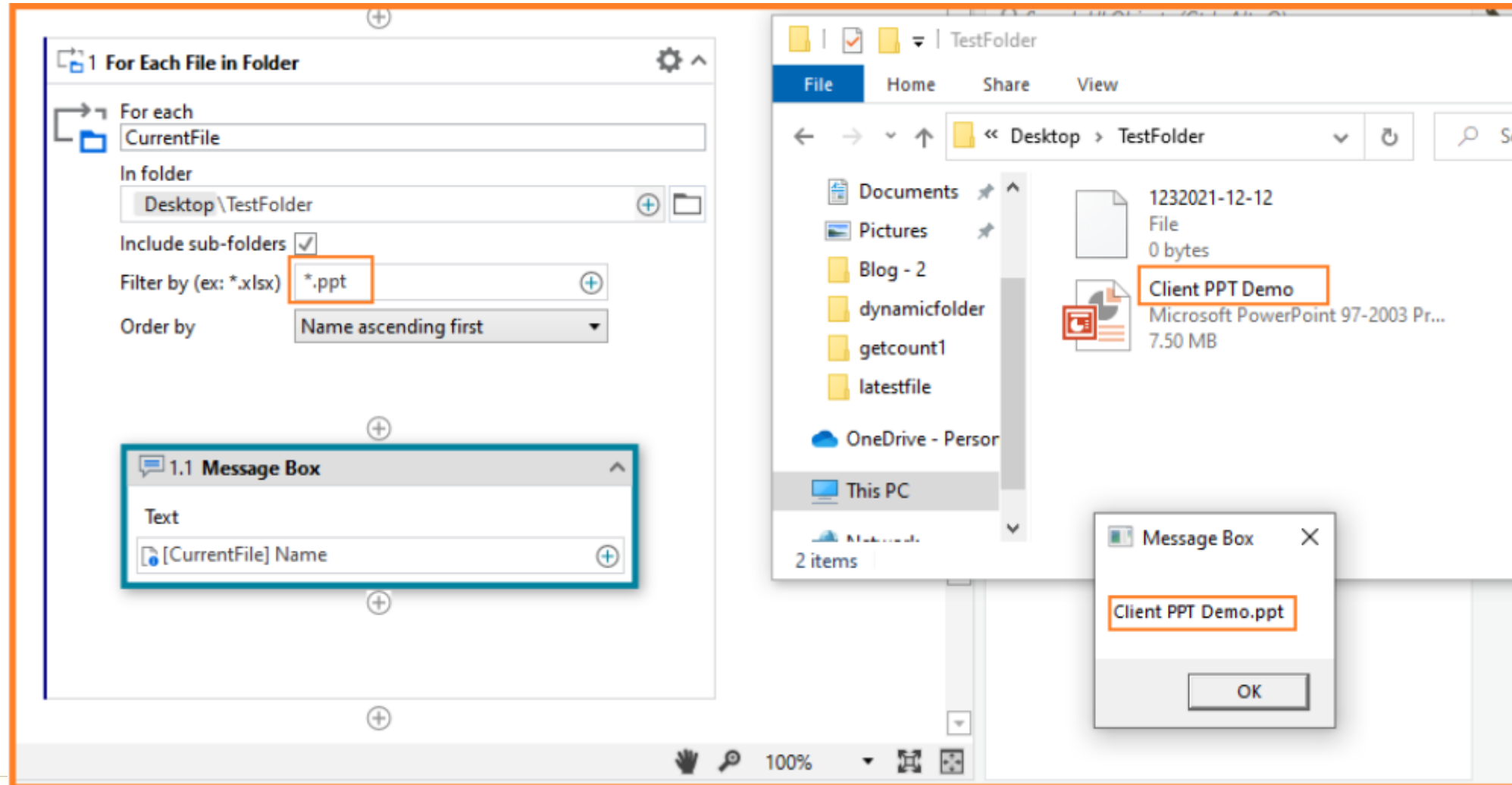
Use 'For Each file in Folder' Activity to loop through all the files in our folder. Select the folder in 'In folder' field and in 'Filter by' field mention. \*.ppt



# Filter files with specific file extension from a folder

- Use message box Activity inside for each loop to display file name in each iteration.

Click run the automation, it will skip other files and will only print files with .ppt extension.



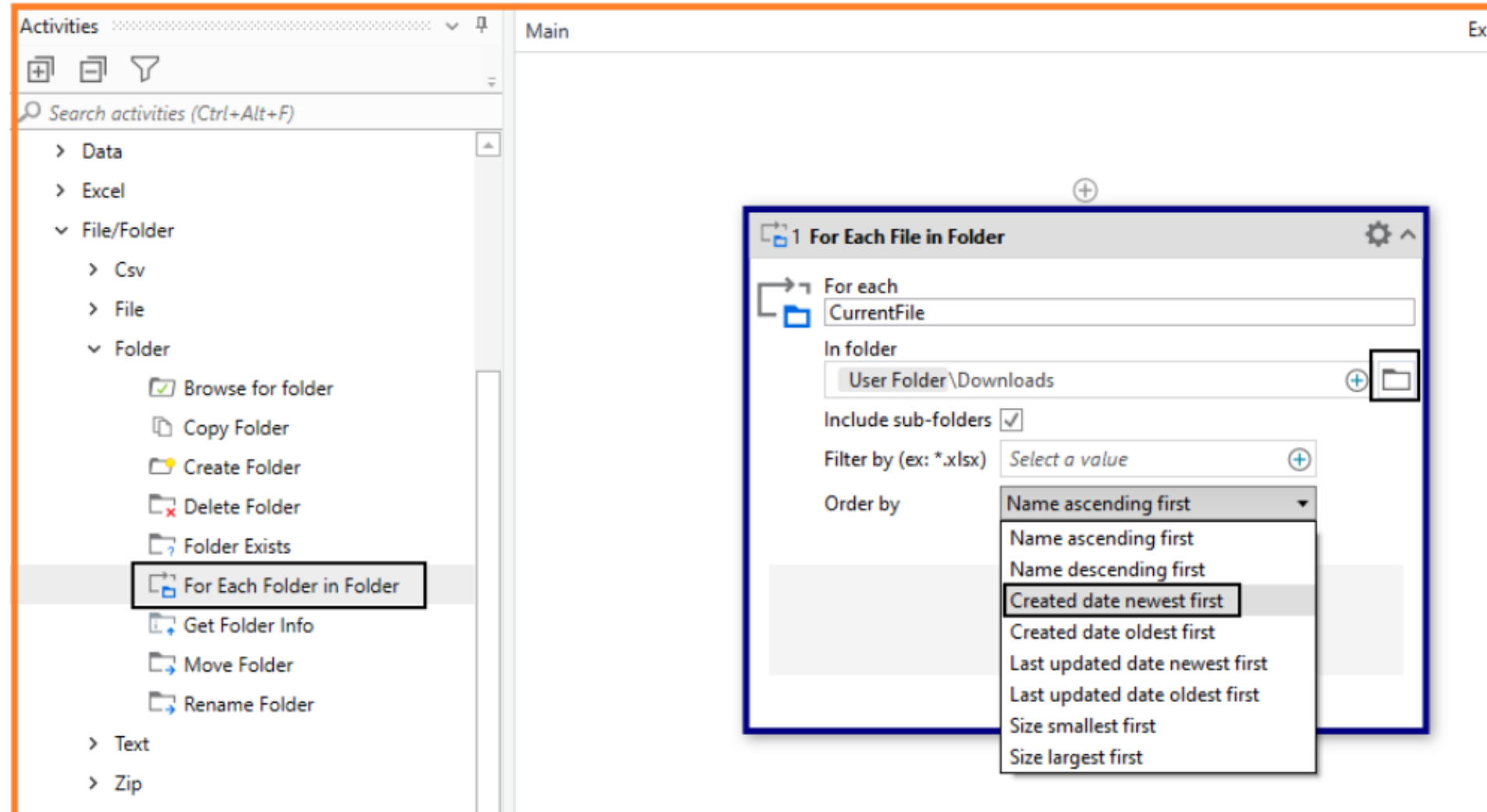
# File and Folder Automation

## Question 3: Get latest file from a specific folder



# Get latest file from a specific folder

Take 'For Each file in Folder' Activity to loop through all the files in download folder. In 'In folder' field, select folder, then in 'Order by' field, select 'created date newest first.'



# Get latest file from a specific folder

Use 'Save For Later' Activity, save the file name as 'LatestFileName' inside the 'For Each file in Folder' Activity.

The screenshot displays the VIPS IDE interface. The main workspace shows two activities:

- 1 For Each File in Folder:** This activity is configured with the following settings:
  - For each:** CurrentFile
  - In folder:** User Folder\Downloads
  - Include sub-folders:** ☒
  - Filter by (ex: \*.xlsx):** Select a value
  - Order by:** Created date newest first
- A\*B 1.1 Save For Later:** This activity is highlighted with an orange border. It has two input fields:
  - Save to:** LatestFileName
  - Value to save:** Select a value

A context menu is open over the 'Value to save' field, showing the following options:

- CurrentFile
- Notebook (Save changes: off)
- Choose Date/Time
- Number
- Text
- Use Saved Value
- Open in Advanced Editor

On the right side, the **Descriptors** panel is visible, showing a list of file descriptors. The **Name** descriptor is highlighted with a black box.

| Descriptors                    |
|--------------------------------|
| Current Index                  |
| Size                           |
| <b>Name</b>                    |
| Full Name (includes full path) |
| Folder                         |
| Last modified date             |
| Last accessed date             |
| Created date                   |
| Extension                      |
| IsReadOnly                     |
| Size in KB                     |

# Get latest file from a specific folder

add the 'Exit loop'  
Activity to break  
further iterations of  
the loop. Now, in the  
message box, print  
the result (Latest file  
name).

The screenshot displays a workflow diagram on the left and a Windows File Explorer window on the right. The workflow diagram, titled '1 For Each File in Folder', is currently 'Running'. It includes a 'For each' loop with 'CurrentFile' as the variable and 'User Folder\Downloads' as the folder. The loop is configured to 'Include sub-folders', 'Filter by (ex: \*.xlsx)' with 'Select a value', and 'Order by' 'Created date newest first'. Inside the loop, there is an activity '1.1 Save For Later' with 'Save to' 'LatestFileName' and 'Value to save' '[CurrentFile] Name'. This is followed by an 'Exit Loop' activity '1.2 Exit Loop'. Below the loop, there is a '2 Message Box' activity with 'Text' 'LatestFileName'. The File Explorer window shows the 'Downloads' folder with a list of files. The file 'final\_2' is highlighted, showing it was modified on '13-12-2021 00:42' and is a 'GIF File'. A message box titled 'Message...' is open, displaying 'final\_2.gif' and an 'OK' button.

| Name                   | Date modified    | Type     |
|------------------------|------------------|----------|
| Today (1)              |                  |          |
| final_2                | 13-12-2021 00:42 | GIF File |
| Last week (1)          |                  |          |
| final_1                | 11-12-2021 21:41 | MP4 File |
| Earlier this month (1) |                  |          |

# File and Folder Automation

**Question 4: Check if a specific folder is empty, or  
find out the number of files in that folder**

# Check if a specific folder is empty, or find out the number of files in that folder

use the '[Get folder Info](#)' Activity to retrieve all of the information of our folder (Desktop\TestFolder) and save it as 'FolderData' for later use.



The screenshot displays the VIPS IDE interface. On the left, a sidebar shows a tree view with 'File' and 'Folder' categories. Under 'Folder', the 'Get Folder Info' activity is highlighted. The main workspace shows a configuration window for '1 Get Folder Info'. The 'Folder path' field is set to 'Desktop\TestFolder'. The 'Output to' field is set to 'Select a value'. A context menu is open over the 'Output to' field, showing options: 'Save for Later Use' and 'Open in Advanced Editor'.

# Check if a specific folder is empty, or find out the number of files in that folder

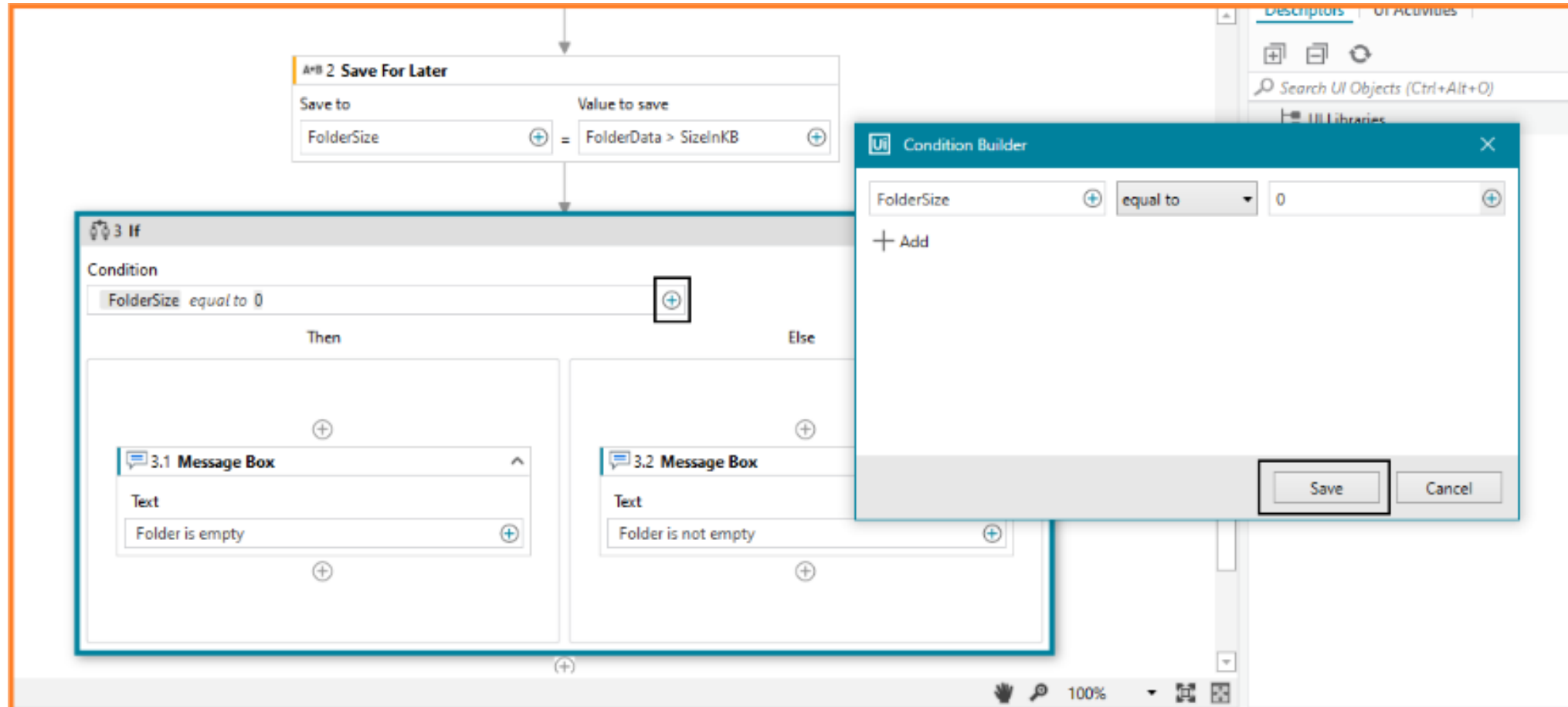
retrieve the size of folder from the information saved above in (FolderData).

(Use Saved Value -> FolderData -> Size in KB)

The screenshot shows a workflow editor with two blocks. The first block, '1 Get Folder Info', has 'Folder path' set to 'Desktop\TestFolder' and 'Output to' set to 'FolderData'. An arrow points from this block to the second block, 'A\*B 2 Save For Later'. The 'Save For Later' block has 'Save to' set to 'FolderSize' and 'Value to save' set to 'Select a value'. A context menu is open over the 'Value to save' field, showing a list of options: 'Files', 'Folders', 'Name', 'Full Path', 'Last modified date', 'Size in KB', 'FolderData', and 'FolderSize'. The 'Size in KB' option is highlighted. The 'FolderData' and 'FolderSize' options are also visible in a separate dropdown menu on the right.

# Check if a specific folder is empty, or find out the number of files in that folder

use the IF activity to check if the size of the folder is 0, for example: if the folder is empty.

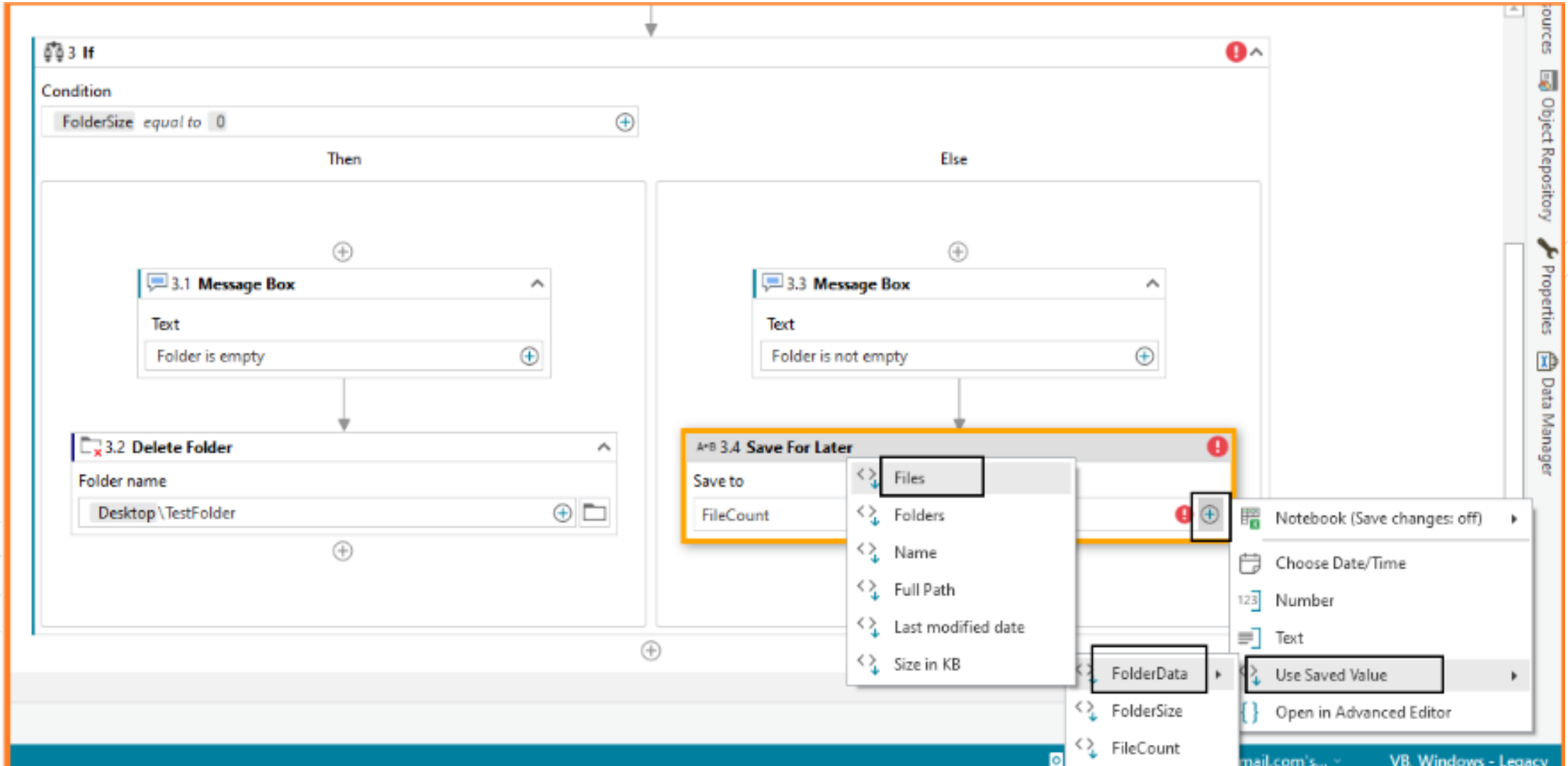


# Check if a specific folder is empty, or find out the number of files in that folder

- Condition will be executed if folder size is zero, and if that is case, delete that folder.
- Include, delete folder Activity in the IF activity's 'Then' section.
- Else condition will be executed if folder size is not zero, i.e., it contains files.
- So, count the number of files in the folder.
- Count of files from the folder information saved above in (FolderData) and save it as 'FileCount' for later use.



# Check if a specific folder is empty, or find out the number of files in that folder



# Check if a specific folder is empty, or find out the number of files in that folder

Print the file count in message box.

The screenshot displays the UiPath Studio interface with a workflow titled "3 If". The workflow is currently in "Running" status. The main canvas shows an "If" activity with the condition "FolderSize equal to 0".

- Then Branch:** Contains a "3.1 Message Box" activity with the text "Folder is empty", followed by a "3.2 Delete Folder" activity with the folder name "Desktop\TestFolder".
- Else Branch:** Contains a "3.3 Message Box" activity with the text "Folder is not empty", followed by a "3.4 Save For Later" activity (labeled "A# 3.4") with "FileCount" as the "Save to" and "FolderData > NoFiles" as the "Value to save". This is followed by a "3.5 Message Box" activity with the text "FileCount".

On the right side, a "TestFolder" window is open, showing a list of files. The file "RPA Fundamentals" is highlighted with an orange box. Below it, a "Message..." dialog box is open, displaying the number "1" in a text field, which is also highlighted with an orange box. The "OK" button is visible at the bottom of the dialog.

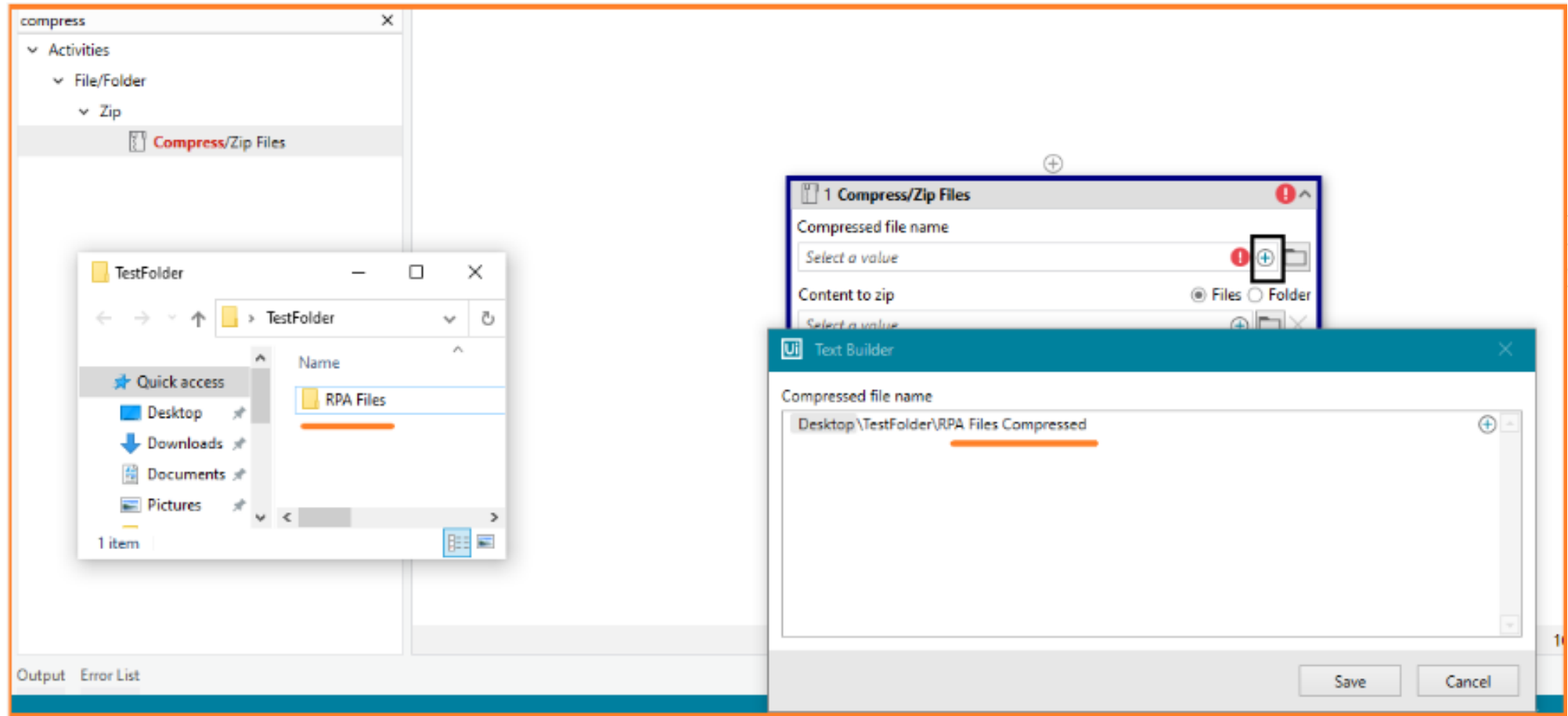
# File and Folder Automation

## Question 5: Zip/Unzip files or folders

# Zip/Unzip files or folders

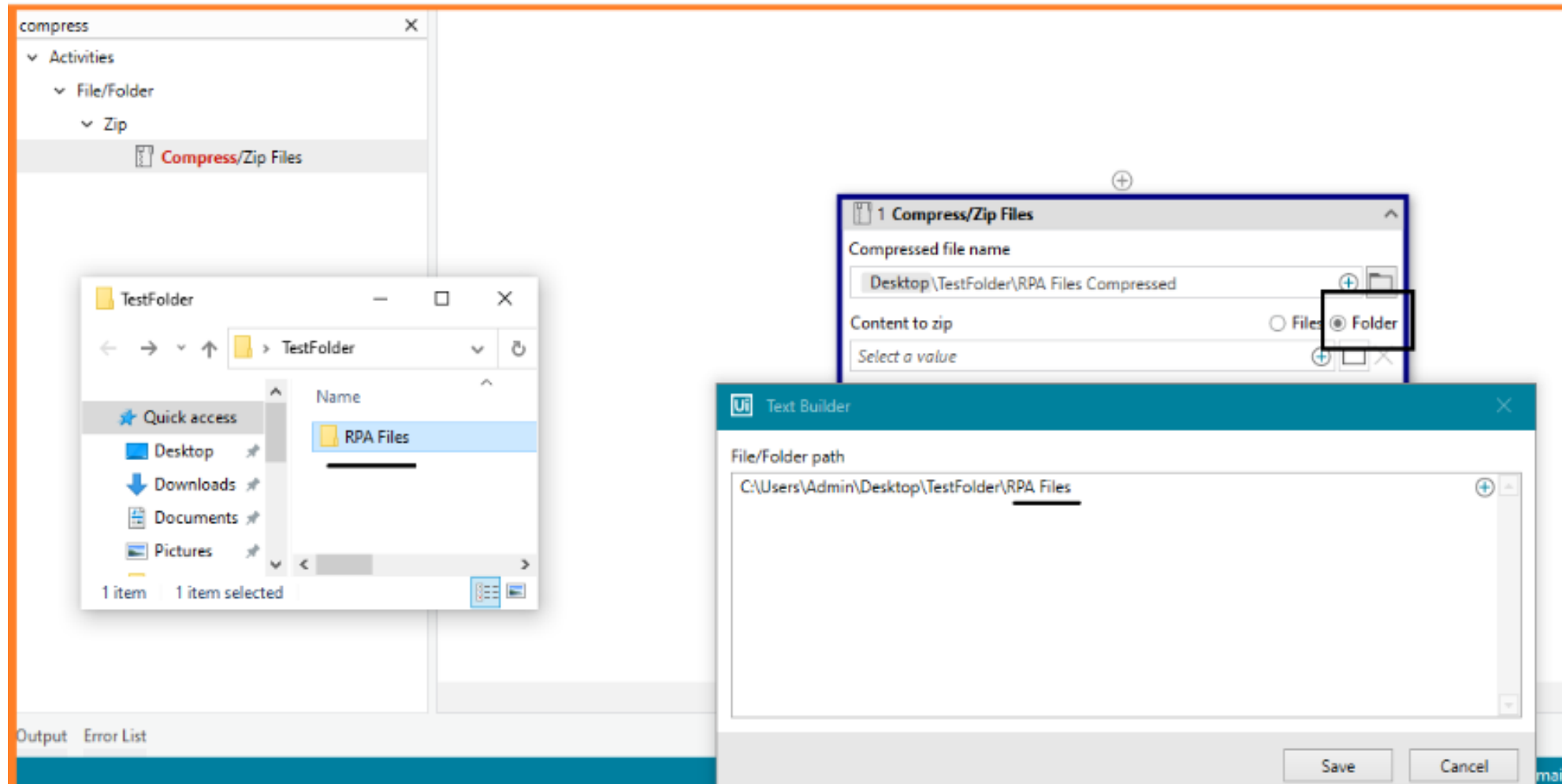
Use  
'[compress/zip files](#)' activity to  
zip folder named  
'rpafiles'.

In 'compressed  
file name' field,  
mention save  
location and  
name of file  
after  
compression



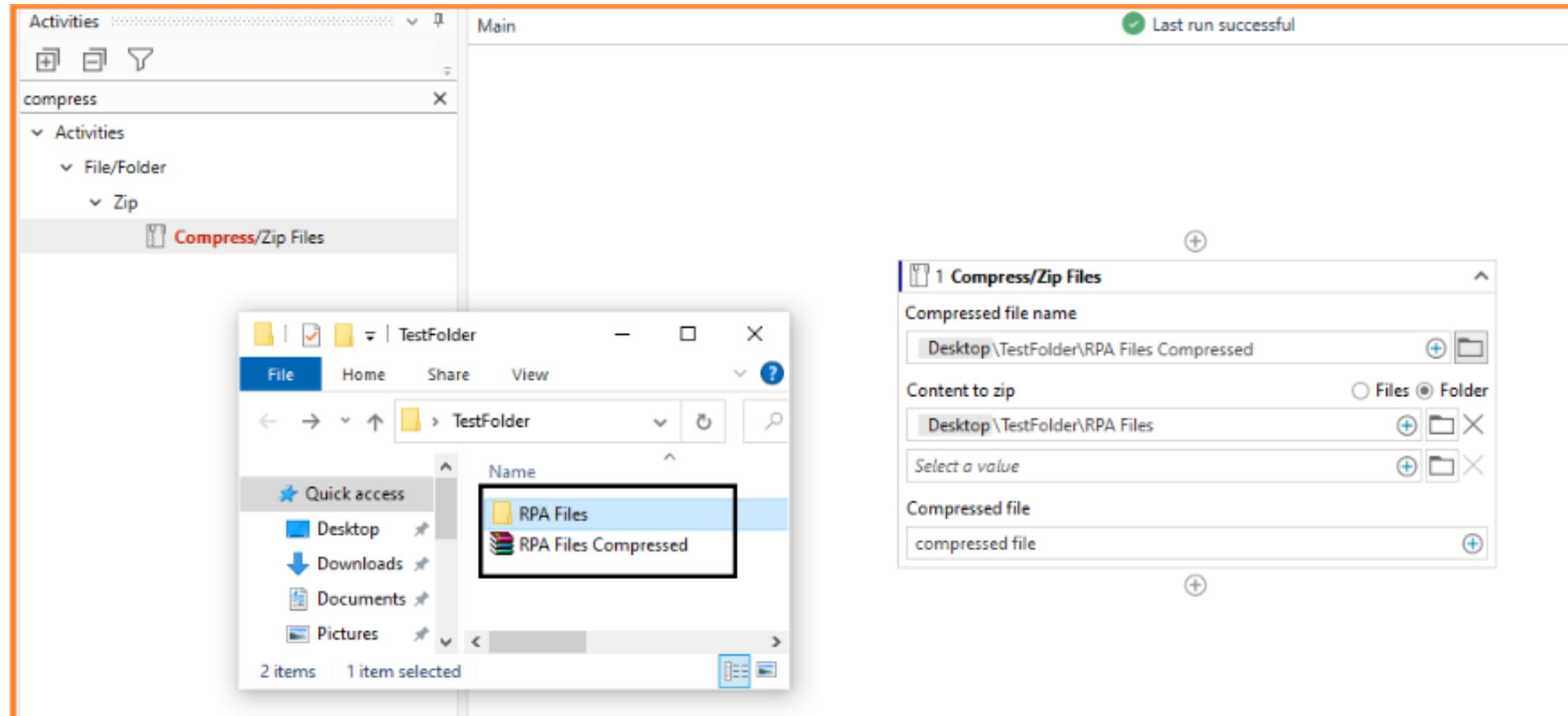
# Zip/Unzip files or folders

Enter full path to folder that needs to be compressed in the 'Content to zip' field (Desktop\TestFolder\RPA Files). Toggle the 'Folder' option as well, as we want to zip the folder.



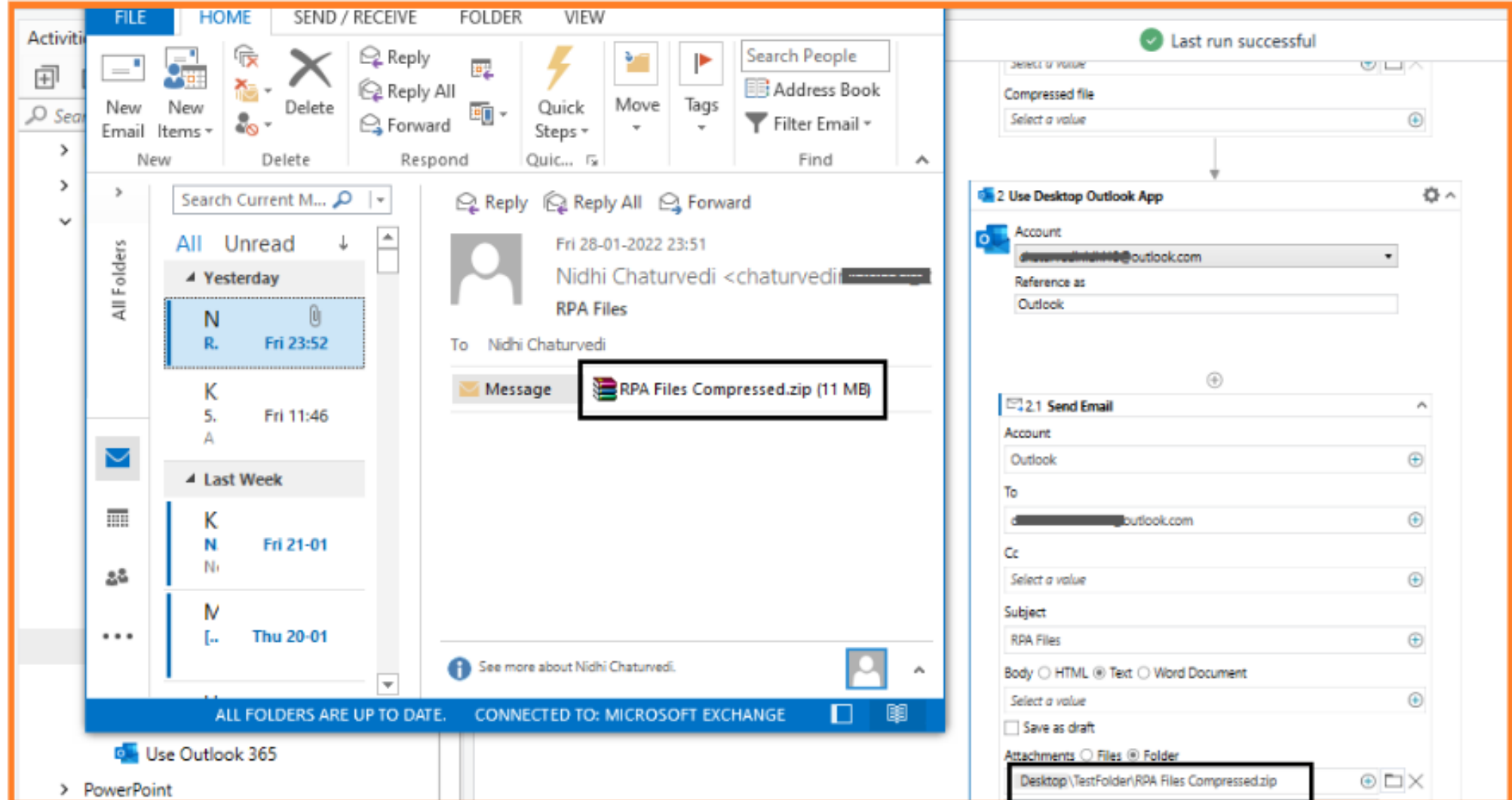
# Zip/Unzip files or folders

run the automation and see if the folder is successfully compressed and placed in the desired location.



# Zip/Unzip files or folders

extend this use case and add send email Activity to send the zipped/compressed file as an attachment in email.



# Syllabus Contents

- **Introduction to UI Automation:** Introduction to UI automation, recording UI interactions, using the Object Repository, UI automation activities, extracting data from an application, using of different control flow activities and its use



# Introduction to UI Automation

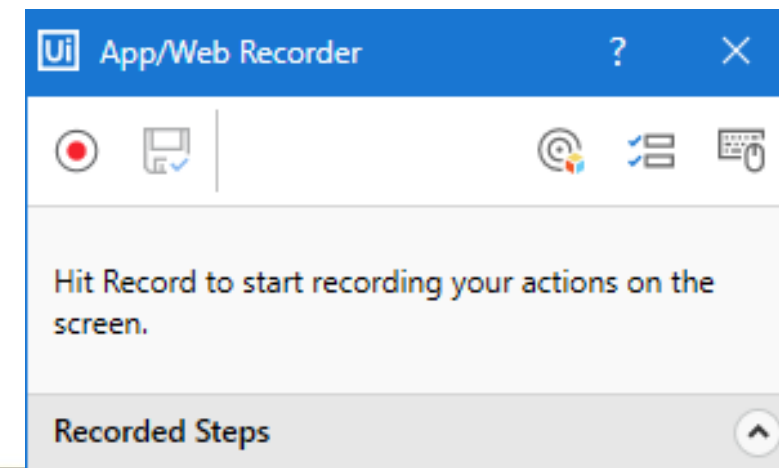
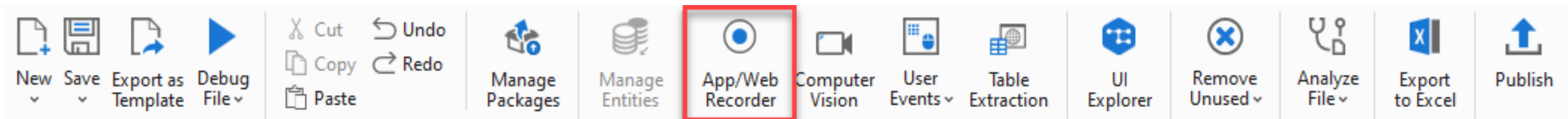
- UI Automation refers to the practice of automating repetitive user interactions with a software application's GUI.
- It allows software tools to mimic human actions like clicking buttons, entering text, and navigating menus.
- Beneficial for:
  - **Automated Testing:** UI automation tools can be used to perform repetitive testing tasks, saving time and ensuring consistency.
  - **Repetitive Tasks:** Automating mundane tasks within an application can significantly improve efficiency.
  - **Regression Testing:** After code changes, UI automation can quickly verify if functionality remains intact.

UI automation is best utilized when Robots and applications run on the same machine because UiPath can integrate directly with the technology behind the application to identify elements, trigger events and get data behind the scenes.

# Key concepts in UI Automation

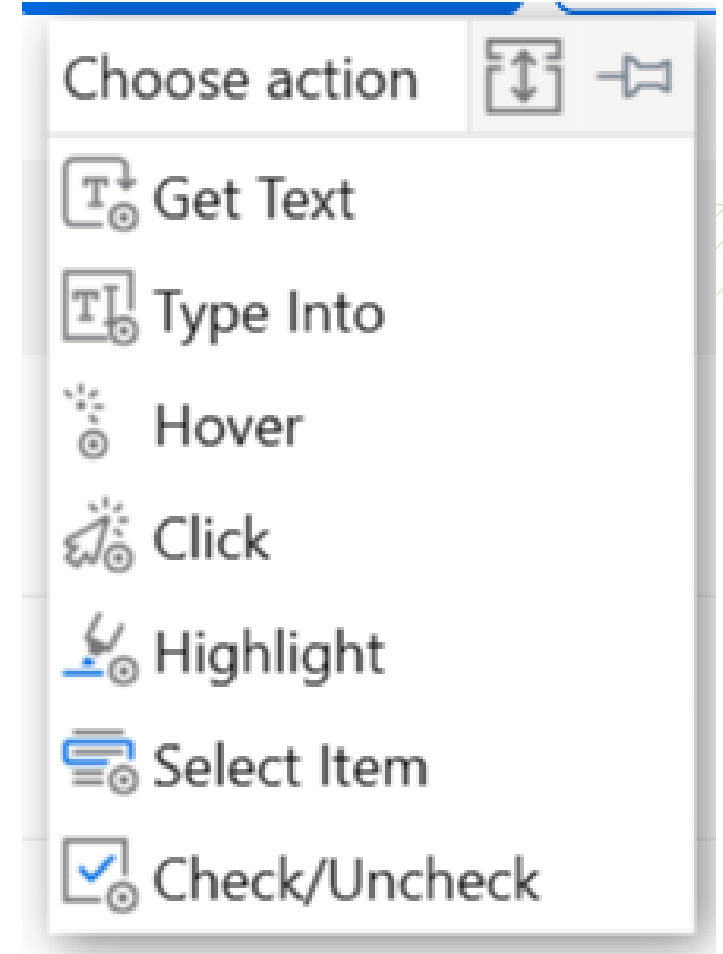
## 1. Recording UI Interactions:

- Many UI automation tools offer recording functionality.
- Allows users to record their actions within the application (clicks, keystrokes, etc.), and the tool translates these actions into scripts that can be replayed.
- Good for beginners, may need refinement for complex scenarios.



## 2. Object Repository:

- Object repository is a central location where UI elements (buttons, text boxes, menus) are identified and stored.
- Acts like a dictionary, associating human-readable names with unique properties that the automation tool uses to interact with the UI elements.
- Simplifies script creation and maintenance, as changes to UI elements only require modification in the object repository.



Browser window showing the URL `https://www.rpasamples.com/unicornname`. The page features a cartoon unicorn robot illustration.

**Your Info**  
**Your Unicorn Name**

What is your name?\*

[Unicorn Name]

What month is your birthday?

**Get Name**

UiApp/Web Recorder interface.

Buttons: Record (red circle), Save (floppy disk), Copy (document with arrow).

Hint: Record and indicate the window you'd like to automate.

Recorded Steps

- **Input Methods**
- Three methods UiPath uses for triggering a **Click** or a **Type Into** activity on an application.
- Displayed as properties in all activities that deal with UI automation.
- SimulateType and SimulateClick Properties
- If **SimulateType** or **SimulateClick** properties are selected, Studio hooks into application and triggers the event handler of an indicated UI element (button, text box).
- Outcome is always dependent of the target technology, such as a web browser or business application.

- Reliable selector should successfully identify same element every time in all conditions, in development, test and production environments and no matter the usernames logged on to applications.
- Tips to improve a selector in the [Selector Editor](#) or [UI Explorer](#):
  - Replace attributes with volatile values with attributes that look steady and meaningful.
  - Replace variable parts of an attribute value with wildcards (\*).
  - If an attribute's value is all wildcard (e.g. name='\*') then the attribute should be removed.
  - If editing attributes doesn't help, try adding more intermediary containers.
  - Avoid using the idx attribute unless its value is a very small number like 1 or 2.

# Selectors

| S  | M  | T  | W  | T  | F  | S  |
|----|----|----|----|----|----|----|
| 25 | 26 | 27 | 28 | 29 | 30 | 1  |
| 2  | 3  | 4  | 5  | 6  | 7  | 8  |
| 9  | 10 | 11 | 12 | 13 | 14 | 15 |
| 16 | 17 | 18 | 19 | 20 | 21 | 22 |
| 23 | 24 | 25 | 26 | 27 | 28 | 29 |
| 30 | 31 | 1  | 2  | 3  | 4  | 5  |

Selector of current UI element is listed below.

```
<html title='Google Calendar - Week of Oct 16, 2016'/>  
<webctrl id='dp_0_tbl' tag='TABLE'/>  
<webctrl id='dp_0_day_23879' tag='TD'/>
```

InitializeFromSelector: 0 ms

| Selector attributes                 |              | Value              |
|-------------------------------------|--------------|--------------------|
| <input type="checkbox"/>            | aname        | 7                  |
| <input type="checkbox"/>            | class        | dp-cell dp-weekday |
| <input type="checkbox"/>            | colName      | F                  |
| <input type="checkbox"/>            | css-selector | body>div>div>div>  |
| <input checked="" type="checkbox"/> | id           | dp_0_day_23879     |
| <input type="checkbox"/>            | innertext    | 7                  |
| <input type="checkbox"/>            | isleaf       | 1                  |



- Selectors can be full or partial (relative).
- Full selectors start with a window or an HTML identifier and have all necessary information to find an element on the whole desktop, while partial selectors work only inside an attach/container that specifies the top-level window where the elements belong:
- **OpenBrowser**
- **OpenApplication**
- **AttachBrowser**
- **AttachWindow**

- Image recognition is last approach to automating applications if nothing else works to identify UI elements on the screen (like selectors or keyboard shortcuts).
- Because image matching requires elements to be fully visible on the screen and that all visible details are the same at runtime as during development, when resorting to image automation extra care should be taken to ensure the reliability of the process.
- Selecting more/less of an image than needed might lead to an image not found or a false-positive match.

- Image matching is sensitive to environment variations like desktop theme or screen resolution.
- When the application runs in Citrix, the resolution should be kept greater or equal than when recording the workflows.
- Otherwise, small image distortions can be compensated by slightly lowering the captured image Accuracy factor.
- Check how the application layout adjusts itself to different resolutions to ensure visual elements proximity, especially in the case of coordinate based techniques like relative click and relative scrape.
- If the automation supports different resolutions, parallel recordings can be placed inside a **PickBranch** activity and the Robot uses either match.

# How to Execute Chrome/Edge/Firefox Automation in PiP Without Interrupting Your Work in the Main Session

- **Conflict with User Data Folders:** By default, Chrome and Edge browsers store user data (history, bookmarks, cookies) in a specific folder created upon first launch. Launching them in separate sessions with the same user data folder creates issues.
- **Custom User Data Folder as a Solution:** Browsers can be launched with a custom user data folder specified by the user. This allows running separate PiP automation sessions without affecting your main browsing session.

- **UiPath Studio Properties for Managing User Data Folders:** UiPath Studio, a Robotic Process Automation (RPA) tool, provides properties within its automation activities to handle user data folders during web automation:
  - **UserDataFolderMode:** This property allows selecting how the browser handles user data folders:
    - UseDefault (default): The activity creates a temporary folder for the automation session.
    - DefaultFolder: The browser's default user data folder is used.
    - UseCustomFolder: A custom user data folder path can be specified using the UserDataFolderPath property.
  - **UserDataFolderPath:** This property allows specifying a custom user data folder path when UserDataFolderMode is set to UseCustomFolder.
- By understanding these points and leveraging the UserDataFolderMode and UserDataFolderPath properties in UiPath Studio, you can effectively run web automation in PiP mode without interfering with your regular browsing activity.

## Create an automation for the following tasks:

- Copy values from an Excel file.
- Paste the values to specific fields in a desktop application.
- Copy the transaction number from the desktop application.
- Paste the value to a cell in the Excel file.
- Click a button in the desktop application to submit the data.
- Display a confirmation message that contains the transaction number and the account number retrieved from the application.

# Setting Up the Project

- **Create a New Project:** Explain how to create a new blank project with default settings.
- **Download Files:** Instruct users to download the provided archive containing the automation project and the necessary files (Transaction.xlsx and DoubleUI.exe).
- **Organize Files:** Guide users on copying Transaction.xlsx to the project folder and opening DoubleUI.exe.

# Adding the Excel File

- **Access Resources Tab:** Show users how to locate the "Resources" tab in the Activities panel.
- **Use Excel File Activity:** Explain the purpose of the "Use Excel File" activity and how to add it to the Designer panel.
- **Configure Excel File:** Instruct users to:
  - Browse and select "Transaction.xlsx".
  - Enter "MyExcel" as the reference name.



# Adding Application and Activities

- **Access Resources Tab:** Remind users to access the "Resources" tab.
- **Use Application/Browser Activity:** Explain the purpose of this activity and how to add it within the "Use Excel File" activity.
- **Indicate Application:** Guide users on how to use "Indicate Application" to select the DoubleUI.exe window.

# Copying Data from Excel to Application

- **Type Into Activity:** Explain the purpose of the "Type Into" activity and how to add it within the "Use Application/Browser" activity.
- **Target Selection:** Instruct users to:
  - Use "Indicate target on screen" to select the "Cash In" field in DoubleUI.exe.
  - Confirm the selection with anchor element (optional).
- **Enter Data from Excel:** Explain how to:
  - Click "+" and select "MyExcel" > "Indicate in Excel".
  - Choose cell B1 (containing the Cash In value) and confirm.

# Copying Data from Excel to Application

- **Repeating Data Entry (2x) Enter Data from Excel** Excel cells B2 and B3 to the "On Us Check" and "Not On Us Check" fields in DoubleUI.exe, respectively
  - Click "+" and select "MyExcel" > "Indicate in Excel".
  - Choose cell B1 (containing the Cash In value) and confirm.

# Retrieving Data from Application to Excel

- **Get Text Activity:** Explain the purpose of the "Get Text" activity and how to add it within the "Use Application/Browser" activity (after the last "Type Into").
- **Target Selection:** Instruct users to:
  - Use "Indicate target on screen" to select the "Transaction #" value in DoubleUI.exe.
  - Confirm the selection with anchor element (optional).
- **Saving Retrieved Data:** Explain how to:
  - Click "+" and select "MyExcel" > "Indicate in Excel".
  - Choose cell B4 and confirm.

# Additional Actions in the Application

- **Click Activity:** Explain the purpose of the "Click" activity and how to add it within the "Use Application/Browser" activity (after "Get Text").
- **Clicking a Button:** Instruct users to:
  - Use "Indicate target on screen" to select the "Accept" button in DoubleUI.exe.
  - Confirm the selection with anchor element (optional).
- **Retrieving Additional Data:** (Optional, repeat steps 1-3 from Slide 7 for another data point, e.g., "Account #")

# Displaying Confirmation Message

- **Access Common Tab:** Guide users to the "Common" tab in the Activities panel.
- **Message Box Activity:** Explain the purpose of the "Message Box" activity and how to add it within the "Use Excel File" activity (after "Use Application/Browser").
- **Building the Message:** Instruct users to:
  - Access the Text Builder.
  - Enter the message template with placeholders for transaction and account numbers.
  - Use "+" to dynamically insert data from cells and saved values.
  - Save the message text.

# Displaying Confirmation Message

- **Running the Automation**
- **Save Project:** Remind users to save the automation project

# Working With UI Automation

1.1 Use Application: DoubleUI

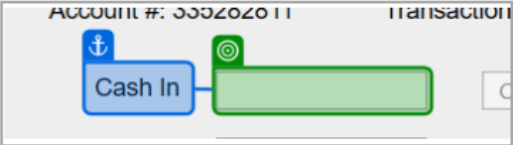


Application path  
{} DoubleUI.exe

Application arguments  
{} Select a value

☐ Match exact title: DoubleUI

1.1.1 Type Into 'Cash In'



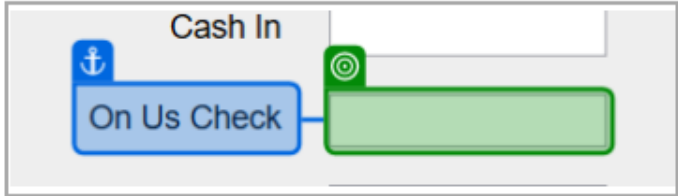
Type this ☒ Standard ☐ Secure  
{} [MyExcel] Sheet1!B1

Empty field before typing Click before typing  
Single line [End, Shift+H] Single

Dr. Divya Agarwal

Single line [End, Shift+H] Single

1.1.2 Type Into 'On Us Check'



Type this ☒ Standard ☐ Secure  
{} [MyExcel] Sheet1!B2

Empty field before typing Click before typing  
Single line [End, Shift+H] Single

1.1.3 Type Into 'Not On Us Check'

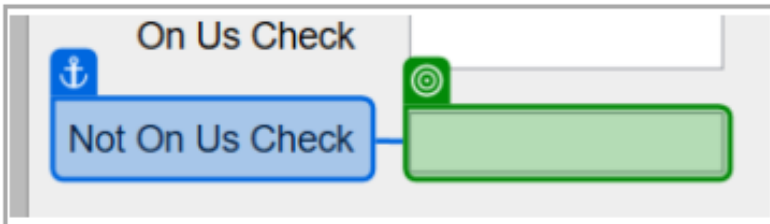




# Working With UI Automation

Single line [End, Shift+H] Single

1.1.3 Type Into 'Not On Us Check'



Type this ☒ Standard ☐ Secure

[MyExcel] Sheet1!B3

Empty field before typing Single line [End, Shift+H]  
Click before typing Single

1.1.4 Get Text 'Transaction #:'

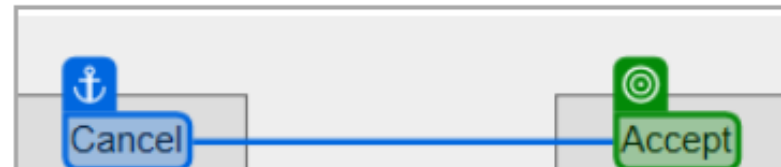
Single line [End, Shift+H] Single

1.1.4 Get Text 'Transaction #:'



Save to  [MyExcel] Sheet1!B4

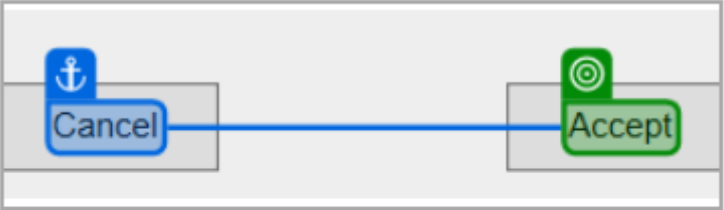
1.1.5 Click 'Accept'



# Working With UI Automation

[myExcel] Sheet1:B4

1.1.5 Click 'Accept'



Click type: Single  
Mouse button: Left

1.1.6 Get Text 'Account #:'

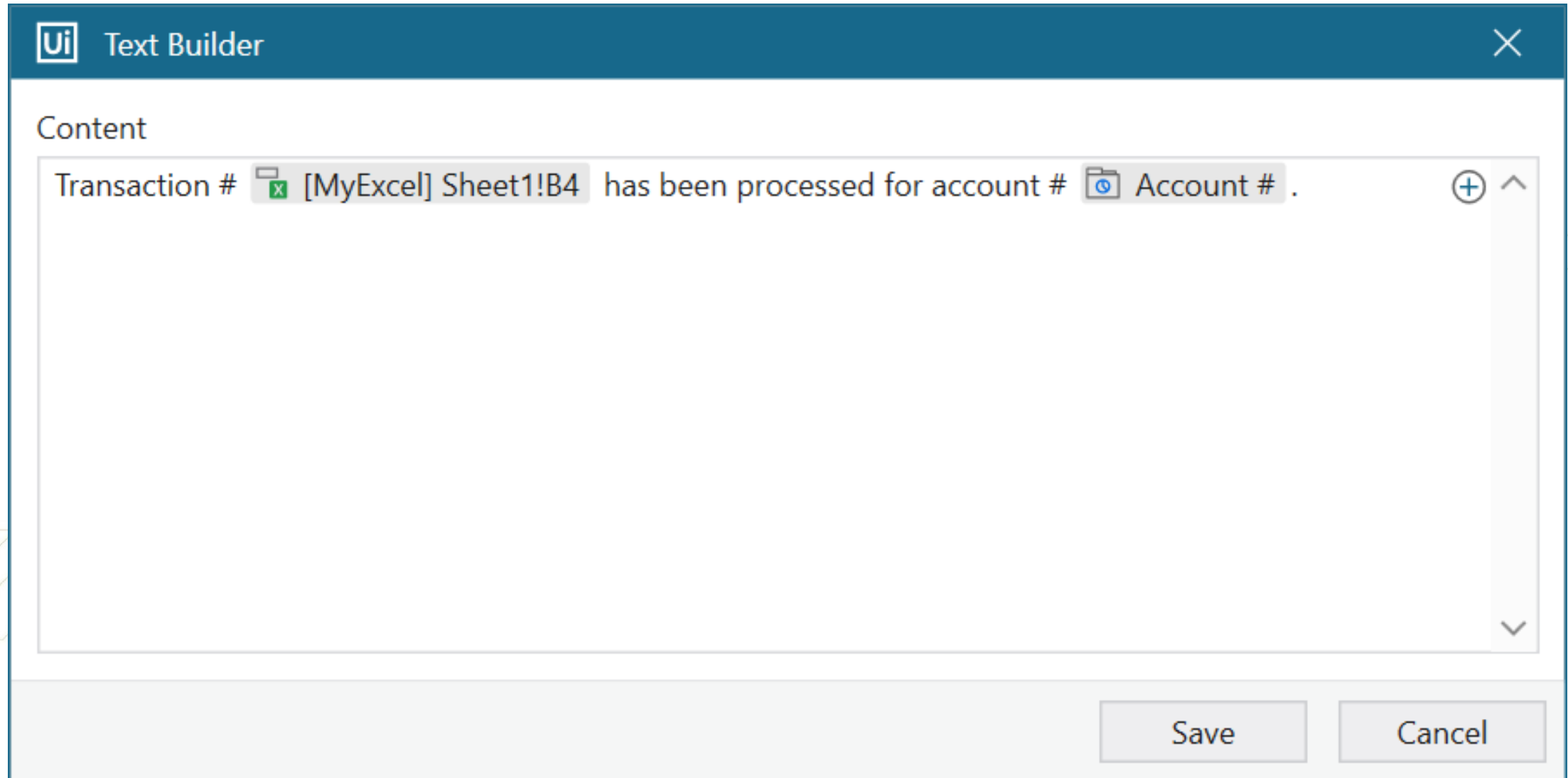
Single Left

1.1.6 Get Text 'Account #:'



Save to: Account #

# Working With UI Automation



# Syllabus Contents

- **UNIT II**

- **Automation using StudioX:** computational concepts when building an automation project, Identify the tasks that are suitable for automation, breakdown a task and document it in a RobotPath. Introduction to file and folder automation,
- **Introduction to UI Automation:** Introduction to UI automation, recording UI interactions, using the Object Repository, UI automation activities, extracting data from an application, using of different control flow activities and its use